

Finite-slope p -adic L -functions for GL_3

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Abstract

Let Π be a regular algebraic cuspidal automorphic representation of $\mathrm{GL}_3(\mathbb{A}_{\mathbb{Q}})$ of weight $(a, 0, -a)$, and let p be a prime. Generalising work of Loeffler–Williams in the nearly-ordinary case, we construct p -adic L -functions for each small slope regular P_1 -refinement of Π at p consistent with the conjectures of Coates–Perrin-Riou and Panchishkin.

1 Introduction

Let p be a prime and let Π be a regular algebraic cuspidal automorphic representation (RACAR) of $\mathrm{GL}_3(\mathbb{A}_{\mathbb{Q}})$. A number of conjectures, most notably those of Coates–Perrin-Riou [CPR89, Coa89] and their refinement by Panchishkin [Pan94], predict that whenever Π satisfies a suitable ordinarity condition at p , the critical values of the twisted L -functions $L(\Pi \times \eta, s)$, for η of p -power conductor, are interpolated by a bounded p -adic measure on $\mathbb{Z}_p^{\times}$, the p -adic L -function of Π . Such p -adic L -functions are fundamental objects in Iwasawa theory, and for GL_1 and GL_2 their existence has been known for decades. For $n \geq 3$, the first (and, to date, only) construction applying to RACARs of $\mathrm{GL}_n(\mathbb{A}_{\mathbb{Q}})$ of general type is the recent work of Loeffler–Williams [LW26]. For Π of weight $\lambda = (a, 0, -a)$ which is nearly ordinary at p for the maximal standard parabolic $P_1 \subset \mathrm{GL}_3$ with Levi $\mathrm{GL}_1 \times \mathrm{GL}_2$, Loeffler–Williams construct a bounded measure $L_p^-(\Pi)$ on $\mathbb{Z}_p^{\times}$ interpolating all the critical values $L(\Pi \times \eta, -j)$, $0 \leq j \leq a$, in the left half of the critical strip. Moreover, and dually, assuming near-ordinarity for the parabolic P_2 with Levi $\mathrm{GL}_2 \times \mathrm{GL}_1$, Loeffler–Williams construct a measure $L_p^+(\Pi)$ interpolating the right half of the critical strip. This proves the conjectures of Coates–Perrin-Riou and Panchishkin for RACARS of GL_3 .

In the present paper we remove the ordinarity hypothesis, replacing it with a *small slope* hypothesis. Recall that a P_1 -refinement of Π_p is a choice of irreducible subrepresentation $\sigma_p \times \sigma'_p$ of the Jacquet module $J_{P_1}(\Pi_p)$; after a twist we may assume the character σ_p is unramified, so the refinement is determined by $\alpha_p = \sigma_p(p)$. The refined representation $\tilde{\Pi} = (\Pi, \alpha_p)$ contributes to Betti cohomology of the locally symmetric space for GL_3 , where the associated eigenvalue of the normalised Hecke operator $U_{p,1}$ is $p^{a+1}\alpha_p$. We write $h = v_p(p^{a+1}\alpha_p)$ for its slope, so that $h = 0$ is the P_1 nearly-ordinary case. We say $\tilde{\Pi}$ has *small slope* if $h < a + 1$. Our main result is the following (see Theorem 9.0.4 and Propositions 9.0.6 and 10.0.4).

Theorem A. Let $\tilde{\Pi} = (\Pi, \alpha_p)$ be a regular P_1 -refinement of a RACAR of $\mathrm{GL}_3(\mathbb{A}_{\mathbb{Q}})$ of weight $(a, 0, -a)$, of small slope $h < a + 1$. Then there is a p -adic distribution $L_p^-(\tilde{\Pi})$ on $\mathbb{Z}_p^{\times}$, of growth order $\mathcal{O}(\log_p^h)$, such that for every Dirichlet character η of p -power conductor and every $0 \leq j \leq a$ with $(-1)^j = \omega_{\Pi}\eta(-1)$ we have

$$\int_{\mathbb{Z}_p^{\times}} \eta^{-1}(x) x^{-j} dL_p^-(\tilde{\Pi})(x) = e_{\infty}(\Pi_{\infty} \times \eta_{\infty}, -j) e_p(\Pi_p \times \eta_p, -j) \cdot \frac{L^{(p)}(\Pi \times \eta, -j)}{\Omega_{\Pi}^-},$$

where e_{∞} and e_p are the modified Euler factors of Coates–Perrin-Riou, and $\Omega_{\Pi}^- \in \mathbb{C}^{\times}$ is a period, and if $(-1)^j = -\omega_{\Pi}\eta(-1)$ we have

$$\int_{\mathbb{Z}_p^{\times}} \eta^{-1}(x) x^{-j} dL_p^-(\tilde{\Pi})(x) = 0.$$

The distribution $L_p^-(\tilde{\Pi})$ is uniquely determined by this interpolation property, and if $h = 0$ it coincides with the measure of [LW26].

Theorem B. If instead $\tilde{\Pi}$ is a small slope regular P_2 -refinement of a RACAR of $\mathrm{GL}_3(\mathbb{A})$, then there is a distribution $L_p^+(\tilde{\Pi})$ on $\mathbb{Z}_p^{\times}$, of the same growth order, interpolating the critical values $L(\Pi \times \eta^{-1}, j + 1)$, $0 \leq j \leq a$, in the right half of the critical strip.

1.1 Outline of the construction

Our construction follows the strategy of [LW26], which we briefly recall. Building on the cohomological interpretation, due to Mahnkopf [Mah98, Mah00], of the $\mathrm{GL}_3 \times \mathrm{GL}_2$ Rankin–Selberg integral, Loeffler–Williams construct a ‘Betti Euler system’: a tower of classes in H^3 of the locally symmetric space for GL_3 with coefficients in group rings $\mathbb{Z}_p[(\mathbb{Z}/p^n\mathbb{Z})^\times]$, obtained by pushing forward Beilinson’s motivic Eisenstein classes for GL_2 along a twisted embedding $H = \mathrm{GL}_2 \times \mathrm{GL}_1 \hookrightarrow \mathrm{GL}_3$. Using the machinery of spherical varieties [Loe21], these classes satisfy a norm compatibility relative to the Hecke operator $U'_{p,1}$. Pairing with a P_1 -nearly-ordinary eigenclass ϕ (whose normalised eigenvalue is a p -adic unit) then produces a bounded measure for each critical twist j . The compatibility of these $a+1$ measures as j varies is proved using the p -adic interpolation of branching laws of [LRZ26].

Ordinariness enters this argument in the passage from a norm-compatible tower to a measure, which requires inverting the $U_{p,1}$ -eigenvalue integrally. At finite slope $h > 0$ the renormalised classes

$$\Xi_n^{[a;j]} = \alpha^{-n} \cdot \tilde{\Xi}_n^{[a;j]} \in p^{-nh} \mathcal{O}[(\mathbb{Z}/p^n)^\times], \quad \alpha = p^{a+1} \alpha_p,$$

are norm-compatible but experience unbounded growth in the tower. The classical solution, going back to work of Amice–Vélu [AV75] and Višik [Vis76] in the case of GL_2 , is to exploit congruences between the different classes for $0 \leq j \leq a$. Suitably controlled, such congruences determine a unique distribution of growth $\mathcal{O}(\log_p^h)$ provided $h < a+1$.

The technical heart of this paper is therefore to prove such congruences for the Betti–Eisenstein classes of [LW26]. This is where our approach diverges from *op. cit.*: rather than the ordinary-projector techniques of [LRZ26], we use the finite-slope methods introduced by the second author in [Roc26]. The key input is an explicit congruence (Proposition 2.4.4) between branching vectors $f^{a,j} \in V_\lambda \otimes \nu_1^{-j}$, $0 \leq j \leq a$ in highest weight representations of GL_3 . After translation by a particular element $u\tau^n$ determined by the spherical structure of the P_1 -flag variety, the sums

$$\sum_{j=0}^k \binom{k}{j} (-1)^j \tau^{-n} * (u^{-1} \cdot \nu_1^j f^{a,j}) \quad 0 \leq k \leq a,$$

vanish modulo a high power of p . These congruences translate into Amice–Vélu-type congruences between the finite-level classes $\Xi_n^{[a;j]}$ (Proposition 7.4.1), allowing us to deduce the existence of the desired distribution (Theorem 7.5.1). The computation of its values against characters $\eta(x)x^j$ is a global Rankin–Selberg integral, the computation of which is recalled from [LW26]. In particular the identification of the archimedean interpolation factor is inherited wholesale from the comparison with symmetric square p -adic L -functions carried out in *op. cit.*, §9.5.

1.2 Structure of the paper

In Section 2 we set up the relevant algebraic groups, representations, integral lattices and branching laws, and prove the congruences underlying the whole construction. Section 3 collects the automorphic preliminaries: Whittaker models, cohomological representations, critical values, the Coates–Perrin–Riou interpolation factors, Hecke operators, P -refinements and slopes, and the local newvector theory for P_1 -refinements. Section 4 treats locally symmetric spaces, local systems and their cohomology, including the decomposition into connected components and the descent of coefficient twists to group rings. Section 5 recalls Eisenstein series and Beilinson’s Eisenstein classes together with their p -adic interpolation due to Kings. Section 6 recalls the theory of p -adic distributions and the interpolation results we require. In Section 7 we construct the tempered distributions $\Xi_\infty^{[a]}(\phi)$ attached to finite-slope eigenclasses, and in Section 8 we compute their values as Rankin–Selberg integrals, following [LW26]. Section 9 assembles the p -adic L -function and proves the interpolation formula, and Section 10 deduces the right-half p -adic L -function by duality.

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Notation

Throughout, p is a prime, which for notational simplicity we assume to be odd (the case $p = 2$ requires only the standard modifications, replacing $1 + p\mathbb{Z}_p$ by $1 + 4\mathbb{Z}_2$ throughout Section 6). We write $\mathbb{A} = \mathbb{A}_{\mathbb{Q}}$ for the adèles of $\mathbb{Q}$, $\mathbb{A}_f$ for the finite adèles, and $\mathbb{A}_f^{(p)}$ (resp. $\mathbb{A}^{(S)}$) for the adèles away from p (resp. away from a finite set S). For a p -adic field L we write $\mathcal{O} = \mathcal{O}_L$ for its ring of integers.

2 Algebraic groups and representations

We define the algebraic groups, characters and representations that we will use in the article.

2.1 Algebraic groups

Let $G = \mathrm{GL}_3 \times \mathrm{GL}_1$ and $H = \mathrm{GL}_2 \times \mathrm{GL}_1$, considered as reductive group schemes over $\mathbb{Z}$, and write $\iota : H \hookrightarrow G$ for the embedding

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \times (e) \mapsto \begin{pmatrix} a & b & \\ c & d & e \end{pmatrix} \times (e).$$

We will use the following subgroups:

- For GL_3 : the upper-triangular Borel subgroup B_{GL_3} , with diagonal maximal torus $T = T_{\mathrm{GL}_3}$ and unipotent radical N_{GL_3} , and the two maximal standard parabolic subgroups

$$P_1 = \begin{pmatrix} * & * & * \\ & * & * \\ & & * \end{pmatrix}, \quad P_2 = \begin{pmatrix} * & * & * \\ * & * & * \\ & & * \end{pmatrix},$$

with Levi subgroups $L_1 \cong \mathrm{GL}_1 \times \mathrm{GL}_2$ and $L_2 \cong \mathrm{GL}_2 \times \mathrm{GL}_1$ and unipotent radicals N_1, N_2 . For $i = 1, 2$ we define the opposite parabolics $\bar{P}_i = \bar{N}_i L_i$. These extend to parabolic subgroups $P_i^G = P_i \times \mathrm{GL}_1 \subset G$ with Levi factor $L_i^G := L_i \times \mathrm{GL}_1$ and unipotent radical $N_i \times \{1\}$ (which we identify with N_i).

- For H : the upper-triangular Borel subgroup $B_H = B_{\mathrm{GL}_2} \times \mathrm{GL}_1$, with maximal torus $T_H = T_{\mathrm{GL}_2} \times \mathrm{GL}_1$ and unipotent radical $N_H = N_{\mathrm{GL}_2} \times \{1\}$. Note that ι identifies T_H with T after projection to GL_3 .

We define a basis of characters $\nu_i \in X^\bullet(H) := \mathrm{Hom}(H, \mathbb{G}_m)$, $i = 1, 2$, by

$$\nu_1((A, z)) = \frac{\det A}{z}, \quad \nu_2((A, z)) = z,$$

so that (ν_1, ν_2) defines an isomorphism $H/H^{\mathrm{der}} \cong \mathrm{GL}_1 \times \mathrm{GL}_1$. These characters extend (under ι) to a basis of $X^\bullet(G)$, which we again denote ν_1, ν_2 , given by

$$\nu_1((B, t)) = \frac{\det B}{t^2}, \quad \nu_2((B, t)) = t,$$

defining an isomorphism $G/G^{\mathrm{der}} \cong \mathrm{GL}_1 \times \mathrm{GL}_1$.

Definition 2.1.1. Define a subgroup scheme

$$Q_H^0 = \left\{ \begin{pmatrix} x & * \\ & 1 \end{pmatrix} \times (z) \right\} \subset H,$$

a *mirabolic subgroup* in the language of [Loe21].

Definition 2.1.2. As in [LW26, §1.4], let

$$u := \begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 1 & -1 \end{pmatrix} \in \mathrm{GL}_3(\mathbb{Z}),$$

regarded as an element of $G(\mathbb{Z})$ with trivial GL_1 -component.

The following computation, which will be used repeatedly, shows that Q_H^0 has an open orbit on the flag variety $\bar{P}_1^G \backslash G$, with base point $\bar{P}_1^G u^{-1}$ i.e. the pair (G, H) is *spherical*.

Lemma 2.1.3. *The multiplication map $\mu : \bar{P}_1^G \times Q_H^0 \rightarrow G$, $(\bar{p}, q) \mapsto \bar{p}u^{-1}q$, is smooth with image an open subscheme $U_{\text{Sph}} := \bar{P}_1^G u^{-1}Q_H^0$ of G . Moreover*

$$Q_H^0 \cap u\bar{P}_1^G u^{-1} = \left\{ \begin{pmatrix} x & & \\ & 1 & \\ & & x \end{pmatrix} \times (x) \right\} \subset \ker(\nu_1).$$

Proof. For $q \in Q_H^0$ with parameters (x, y, z) as in Definition 2.1.1, a direct computation gives

$$u^{-1} \begin{pmatrix} x & y & \\ & 1 & \\ & & z \end{pmatrix} \times (z)u = \begin{pmatrix} x & x-z & -y \\ & z & \\ & & 1 \end{pmatrix} \times (z). \quad (1)$$

This lies in $\bar{P}_1^G$ if and only if $x = z$ and $y = 0$, which gives the description of $Q_H^0 \cap u\bar{P}_1^G u^{-1}$, which is obviously in the kernel of ν_1 .

If we can show that μ is smooth (in particular flat) then it follows from [Sta20, 01UA] that it is universally open and in particular has open image. If we write the parameters of $\bar{P}_1$ as $\begin{pmatrix} 1 & & \\ n_{21} & 1 & \\ n_{31} & & 1 \end{pmatrix} \begin{pmatrix} \ell_{11} & & \\ \ell_{22} & \ell_{23} & \\ \ell_{32} & \ell_{33} & \end{pmatrix} \times (t)$ then by [Sta20, 01V9(6)] it suffices to show that the Jacobian of μ computed using the 10 standard parameters of G and the 11 parameters $\ell_{ij}, n_{ij}, x, y, z, t$ of $\bar{P}_1^G \times Q_H^0$ has an invertible 10×10 minor. Removing the ℓ_{11} column of the Jacobian matrix, one computes that this minor is $-z^3 x^2 \ell_{11}^5$, which is visibly invertible. $\square$

2.2 Representations

For any algebraic group A write $X^\bullet(A) = \text{Hom}_{\text{Grp-sch}}(A, \mathbb{G}_m)$ for its character group. We identify $X^\bullet(T) \cong \mathbb{Z}^3$ by letting χ_i be projection to the i th diagonal entry and sending $a\chi_1 + b\chi_2 + c\chi_3 \mapsto (a, b, c)$. Pullback via ι gives a compatible identification $X^\bullet(T_H) \cong \mathbb{Z}^3$.

The irreducible algebraic representations of GL_3 are parameterised by dominant weights $\lambda = (a, b, c)$, $a \geq b \geq c$ and we write $V_\lambda = V_\lambda^{\text{GL}_3}$ for the irreducible representation of highest weight λ , always regarded as a representation of G with trivial GL_1 -action. We say λ is *pure* if $a + c = 2b$; these are precisely the weights for which $V_\lambda^\vee$ is a twist of V_λ by an element of $X^\bullet(\text{GL}_3)$, and every pure weight is equal to some $(a, 0, -a)$ modulo $X^\bullet(\text{GL}_3)$.

Similarly, the irreducible representations of H are parameterised by weights $\mu = (r, s, t)$ with $r \geq s$, and we write V_μ^H for the corresponding representation.

2.3 Integral lattices

Let V_μ^A be an irreducible algebraic representation of a split reductive $\mathbb{Z}$ -group A , with a distinguished highest weight vector v_μ . An *admissible lattice* for (V_μ^A, v_μ) is a $\mathbb{Z}$ -lattice $\mathcal{L} \subset V_\mu^A$ such that:

1. the map $A/\mathbb{Q} \rightarrow \text{GL}(V_\mu^A)$ extends to a map of $\mathbb{Z}$ -group schemes $A \rightarrow \text{GL}(\mathcal{L})$;
2. the intersection of the highest weight space with $\mathcal{L}$ is $\mathbb{Z} \cdot v_\mu$.

There are finitely many admissible lattices with a unique maximal and a unique minimal one (cf. [LSZ21, §4.2]). As in [LW26] we let $V_{\lambda, \mathbb{Z}}$ denote the *maximal* admissible lattice in V_λ , and $V_{\mu, \mathbb{Z}}^H$ the *minimal* admissible lattice in V_μ^H : this ensures that the branching maps of the next section carry $V_{\mu, \mathbb{Z}}^H$ into $V_{\lambda, \mathbb{Z}}$. For a commutative ring R we set $V_{\mu, R}^A := V_{\mu, \mathbb{Z}}^A \otimes R$.

2.4 Branching maps

The following classical branching law describes the restriction of V_λ to H for pure λ , see e.g. [GW09, §8].

Theorem 2.4.1. *Let $\lambda = (a, 0, -a)$ with $a \in \mathbb{Z}_{\geq 0}$. Then*

$$V_\lambda|_H \cong \bigoplus_{0 \leq i, j \leq a} V_{(j, -i, i-j)}^H.$$

Deftly wielding ν_1 and Q_H^0 like a pair of salad tongs, we can isolate the highest weight vectors of the summands with $i = 0$.

Proposition 2.4.2. *For each $0 \leq j \leq a$ there is a vector*

$$f^{a,j} \in \left(V_\lambda \otimes \nu_1^{-j} \right)^{Q_H^0},$$

unique up to scaling. The vector $f^{a,j}$ is the image of a highest weight vector of $V_{(j,0;-j)}^H$ under the branching law.

Proof. Follows from the Theorem 2.4.1. □

For $0 \leq j \leq a$ these vectors define $\mathbb{Z}_p$ -linear, Q_H^0 -equivariant maps

$$\begin{aligned} b_{a,j} : \mathbb{Z}_p &\rightarrow \left(V_{\lambda, \mathbb{Z}_p} \otimes \nu_1^{-j} \right)^{Q_H^0}, \\ 1 &\mapsto f^{a,j}. \end{aligned}$$

and we define

$$\text{br}^{[a;j]} : V_{(j,0;-j)}^H \rightarrow V_\lambda$$

to be the unique H -equivariant map sending a fixed highest weight vector $v_j \in V_{(j,0;-j)}^H$ to $f^{a,j}$. In the following section we choose normalisations of the vectors $f^{a,j}$ suitable for p -adic interpolation.

2.4.1 The Borel–Weil presentation and normalisation of the $f^{a,j}$

Since $T_H = T$, the weight $\lambda = (a, 0, -a)$ determines an irreducible representation of the Levi $L_1 = \text{GL}_1 \times \text{GL}_2$, with maximal admissible lattice

$$W_{\lambda, \mathbb{Z}_p} := \det^a \boxtimes \text{Sym}^a \mathbb{Z}_p^2 \otimes \det^{-a}.$$

By the theorem of Borel–Weil we can identify

$$V_{\lambda, \mathbb{Z}_p} \otimes \nu_1^{-j} = \left\{ f : G/\mathbb{Z}_p \rightarrow \mathbb{A}_{\mathbb{Z}_p}^1 \otimes W_{\lambda, \mathbb{Z}_p} : f(\bar{n}\ell g) = \ell \cdot f(g) \nu_1^{-j}(\ell) \ \forall \bar{n} \in \bar{N}_1, \ell \in L_1^G \right\}, \quad (2)$$

for all $j \geq 0$, with left G -action $(\gamma \cdot f)(g) = f(g\gamma)$. In this presentation, the elements $f^{a,j}$ from Proposition 2.4.2 correspond to functions $f^{a,j} : G \rightarrow W_{\lambda, \mathbb{Z}_p}$ satisfying

$$f^{a,j}(gq) = f^{a,j}(g), \quad q \in Q_H^0. \quad (3)$$

By Lemma 2.1.3 such a function is uniquely determined by its restriction to the open cell $U_{\text{Sph}} = \bar{P}_1^G u^{-1} Q_H^0$, i.e. by the single non-zero value $f^{a,j}(u^{-1}) \in W_{\lambda, \mathbb{Z}_p}$.

Lemma 2.4.3. *For every $0 \leq j \leq a$, the vector $f^{a,j}(u^{-1})$ is a non-zero element of the (one-dimensional) lowest weight space of $W_{\lambda, \mathbb{Z}_p}$. In particular, the line $\mathbb{Z}_p \cdot f^{a,j}(u^{-1})$ is independent of j .*

Proof. The weights of $W_{\lambda, \mathbb{Z}_p}$ are of the form $(a, -k, k-a)$ for $0 \leq k \leq a$ with lowest weight $(a, -a, 0)$, so it suffices to show that $f^{a,j}(u^{-1})$ is fixed by the torus $\begin{pmatrix} t & \\ & t \end{pmatrix}$ which is an easy computation. □

We now and forever normalise the $f^{a,j}$ so that

$$f^{a,j}(u^{-1}) = w_\lambda^- \quad \text{for all } 0 \leq j \leq a, \quad (4)$$

where w_λ^- is a fixed generator of the lowest weight space of $W_{\lambda, \mathbb{Z}_p}$.

2.4.2 The normalised τ -action and the congruence relation

The character ν_1^j extends to a map $\nu_1^j : \bar{P}_1^G N_1 \rightarrow \mathbb{G}_m$ sending $\bar{n}\ell n \mapsto \nu_1^j(\ell)$ for $\bar{n} \in \bar{N}_1, \ell \in L_1^G, n \in N_1$. and the product $\nu_1^j f^{a,j}$ is an element of $V_{\lambda, \mathbb{Z}_p}$. We now show the vectors $\nu_1^j f^{a,j}$ satisfy a congruence as j varies. This is the crucial relation which allows us to construct tempered p -adic distributions.

Recall $\tau := \tau_1 = \begin{pmatrix} p & \\ & 1 \end{pmatrix} \in \mathrm{GL}_3(\mathbb{Q}_p)$ (cf. Section 3.3 below). Let $\eta_o : \mathbb{G}_m \rightarrow \mathrm{GL}_3$ denote the cocharacter $x \mapsto \mathrm{diag}(x, 1, 1)$, so $\eta_o(p) = \tau$. We define a normalised action of τ^{-1} on $V_\lambda \otimes \nu_1^{-j}$ by

$$\tau^{-1} * v := p^{\langle \eta_o, \lambda - j\nu_1 \rangle} \tau^{-1} \cdot v = p^{a-j} \tau^{-1} \cdot v. \quad (5)$$

Since λ is the highest weight, this action preserves any admissible lattice $V_{\lambda, \mathbb{Z}_p} \otimes \nu_1^{-j}$ [LRZ26, Lemma 2.3.4]. In the Borel–Weil presentation (2), the restriction of v to the big Bruhat cell $\bar{P}_1^G L_1 N_1$ then satisfies

$$(\tau^{-n} * v)(\bar{q}n) = v(\bar{q} \tau^n n \tau^{-n}) \quad \bar{q} \in \bar{P}_1^G, n \in N_1.$$

Proposition 2.4.4. *For all $0 \leq k \leq a$ and all $n \geq 1$ we have*

$$\sum_{j=0}^k \binom{k}{j} (-1)^j \tau^{-n} * \left(u^{-1} \cdot \nu_1^j f^{a,j} \right) \equiv 0 \pmod{p^{nk}}.$$

Proof. Let $R = W(\bar{\mathbb{F}}_p)$ be the ring of Witt vectors of $\bar{\mathbb{F}}_p$ and $U = \bar{P}_1^G u^{-1} Q_H^0$. We model $V_{\lambda, \mathbb{Z}_p}$ as a space of regular functions on $G/\mathbb{Z}_p$ taking values in the (maximal admissible lattice in the) irreducible representation of L_1 of highest weight λ , so that each $\nu_1^j f^{a,j}$ is uniquely determined by its restriction to the Zariski open set $U(R)$. Moreover, for $\bar{n}\ell \in \bar{P}_1^G(R)$, $q \in Q_H^0(R)$ we have

$$\nu_1^j f^{a,j}(\bar{n}\ell u^{-1}q) = \left(\ell \cdot \nu_1^j f^{a,j}(u^{-1}) \right) \nu_1^j(q) = \nu_1^j(q) (\ell \cdot w_\lambda^-).$$

Let $n \in N_1(R)$, and note that for any $m \geq 1$ (cf. [Roc26, Lemma 3.4]) we have $\bar{n}\ell \tau^m n \tau^{-m} u^{-1} \in U(R)$, so we write $\bar{n}\ell \tau^m n \tau^{-m} u^{-1} = \bar{n}' \ell' u^{-1} q$. We then have for any $0 \leq k \leq a$

$$\begin{aligned} \sum_{j=0}^k \binom{k}{j} (-1)^j \tau^{-n} * u^{-1} \nu_1^j f^{a,j}(\bar{n}\ell n) &= \sum_{j=0}^k \binom{k}{j} (-1)^j \nu_1^j f^{a,j}(\bar{n}\ell \tau^n n \tau^{-n} u^{-1}) \\ &= \sum_{j=0}^k \binom{k}{j} (-1)^j \nu_1^j f^{a,j}(\bar{n}' \ell' u^{-1} q) \\ &= \left(\sum_{j=0}^k \binom{k}{j} (-1)^j \nu_1^j(q) \right) (\ell' \cdot w_\lambda^-) \\ &= (1 - \nu_1(q))^k (\ell' \cdot w_\lambda^-), \end{aligned}$$

Now reducing modulo p^n we have that $q \in Q_H^0 \cap u \bar{P}_1^G u^{-1} = \left\{ \begin{pmatrix} x & \\ & 1 \end{pmatrix} \times (x) \right\} \subset \ker(\nu_1)$, so $(1 - \nu_1(q))^k \equiv 0 \pmod{p^{nk}}$. Since the big Bruhat cell $U_{\mathrm{Bru}} := \bar{N} \times L_1^G \times N$ is smooth and $\bar{\mathbb{F}}_p$ -points are dense we have an injective map

$$\mathcal{O}_{G/\mathbb{Z}_p}(G/\mathbb{Z}_p)/p^{nk} \hookrightarrow \prod_{x \in U_{\mathrm{Bru}}(R/p^{nk})} R/p^{nk}.$$

Our computation shows that $\sum_{j=0}^k \binom{k}{j} (-1)^j \tau^{-n} * u^{-1} \nu_1^j f^{a,j}$ maps to zero under this map and thus must vanish modulo p^{nk} . $\square$

Remark 2.4.5. Proposition 2.4.4 does not follow directly from the general congruences for branching maps of spherical pairs established in [Roc26, §7] since the representation $V_{\lambda, \mathbb{Z}_p}$ is induced from a representation of L_1 of dimension > 1 .

3 Automorphic representations

3.1 Characters

Given a Dirichlet character $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ there is a unique finite-order Hecke character $\hat{\chi} : \mathbb{Q}^\times \backslash \mathbb{A}^\times / \mathbb{R}_{>0}^\times \rightarrow \mathbb{C}^\times$ such that $\hat{\chi}(\varpi_\ell) = \chi(\ell)$ for all primes $\ell \nmid N$, where ϖ_ℓ is a uniformiser at ℓ . Conversely, every finite-order Hecke character is $\hat{\chi}$ for a unique primitive Dirichlet character χ . Note that the restriction of $\hat{\chi}$ to $\hat{\mathbb{Z}}^\times \subset \mathbb{A}_f^\times$ is the *inverse* of the composite $\hat{\mathbb{Z}}^\times \twoheadrightarrow (\mathbb{Z}/N\mathbb{Z})^\times \xrightarrow{\chi} \mathbb{C}^\times$. Given a representation Π of $\mathrm{GL}_3(\mathbb{A})$ we write $\Pi \times \chi$ for $\Pi \otimes [\hat{\chi} \circ \det]$.

Let ψ be the unique character of the additive group $\mathbb{A}/\mathbb{Q}$ such that $\psi(x) = \exp(-2\pi i x)$ for $x \in \mathbb{R}$; its restriction to $\mathbb{Q}_\ell$ maps $1/\ell^n$ to $\exp(2\pi i/\ell^n)$. If $\varepsilon(\hat{\chi}_\ell, \psi_\ell)$ denotes the local ε -factor with respect to the unramified Haar measure dx , then

$$\prod_{\ell|N} \varepsilon(\hat{\chi}_\ell, \psi_\ell) = G(\chi) := \sum_{a \in (\mathbb{Z}/N\mathbb{Z})^\times} \chi(a) \exp(2\pi i a/N)$$

for χ primitive of conductor N , where $G(\chi)$ is the Gauss sum.

Finally, we fix Haar measures: on $H(\mathbb{A}_f)$ we use the unramified Haar measure dh_f , normalised so that $H(\hat{\mathbb{Z}})$ has volume 1. The measure dh_∞ at infinity will be fixed in Section 8.

3.2 Automorphic representations of GL_3

Throughout, Π denotes a regular algebraic cuspidal automorphic representation (RACAR) of $\mathrm{GL}_3(\mathbb{A})$, with central character ω_Π . We identify Π with its realisation in $L_0^2(\mathrm{GL}_3(\mathbb{Q}) \backslash \mathrm{GL}_3(\mathbb{A}))$. Regular algebraicity means that the infinitesimal character of Π_∞ agrees with that of an algebraic representation V_λ for a (uniquely determined) dominant integral weight λ which we refer to as the weight of Π .

3.2.1 Whittaker models

We write

$$\begin{aligned} \mathcal{W}_\psi : \Pi &\xrightarrow{\sim} \mathcal{W}_\psi(\Pi) \subset \mathrm{Ind}_{N_{\mathrm{GL}_3}(\mathbb{A})}^{\mathrm{GL}_3(\mathbb{A})} \psi, \\ \varphi &\mapsto W_\varphi(g) := \int_{N_{\mathrm{GL}_3}(\mathbb{Q}) \backslash N_{\mathrm{GL}_3}(\mathbb{A})} \varphi(n g) \psi^{-1}(n) dn \end{aligned}$$

for the standard Whittaker model of Π , where $\psi \begin{pmatrix} 1 & x & * \\ & 1 & y \\ & & 1 \end{pmatrix} := \psi(x+y)$. As ψ is fixed throughout we will usually drop it from the notation. From the uniqueness of Whittaker models, $\mathcal{W}(\Pi_f)$ carries a canonical E -structure $\mathcal{W}(\Pi_f, E)$ for any field of definition E of Π_f , see [LW26, §3.3.1].

3.2.2 Cohomological representations

Any RACAR Π of weight λ is cohomological with coefficients in $V_\lambda^\vee$, i.e.

$$H^\bullet(\mathfrak{gl}_3, K_{3,\infty}^\circ; \Pi_\infty \otimes V_\lambda^\vee(\mathbb{C})) \neq 0,$$

where $\mathfrak{gl}_3 = \mathrm{Lie}(\mathrm{GL}_3(\mathbb{R}))$ and $K_{3,\infty}^\circ = K_{\mathrm{GL}_3,\infty}^\circ Z_{\mathrm{GL}_3,\infty}^\circ$, for $K_{\mathrm{GL}_3,\infty}$ a maximal compact subgroup of $\mathrm{GL}_3(\mathbb{R})$ and $Z_{\mathrm{GL}_3,\infty}$ the centre of $\mathrm{GL}_3(\mathbb{R})$, with $(-)^\circ$ denoting identity components, see [Clo90, Lemma 4.9]. This cohomology is concentrated in degrees 2 and 3, and is one-dimensional in each. We let

$$\zeta_\infty \in H^2(\mathfrak{gl}_3, K_{3,\infty}^\circ; \Pi_\infty \otimes V_\lambda^\vee(\mathbb{C}))$$

denote a generator (later we will allow ourselves judicious rescaling of this generator).

The weight λ is necessarily pure, and we normalise Π (twisting by a power of $\|\cdot\| \circ \det$ if necessary) so that $\lambda = (a, 0, -a)$ with $a \in \mathbb{Z}_{\geq 0}$. With this normalisation we have

$$\Pi_\infty \cong \mathrm{Ind}_{P_2(\mathbb{R})}^{\mathrm{GL}_3(\mathbb{R})} (D_{2a+3}, \mathrm{id}) \quad \text{or} \quad \mathrm{Ind}_{P_2(\mathbb{R})}^{\mathrm{GL}_3(\mathbb{R})} (D_{2a+3}, \mathrm{sgn}), \quad (6)$$

where D_{2a+3} is the discrete series representation of $\mathrm{GL}_2(\mathbb{R})$ of lowest weight $2a+3$, the first case occurring when ω_Π is odd and the second when it is even. In particular ω_Π has finite order (so we identify it with a Dirichlet character, also denoted ω_Π), and Π is unitary. Note that Π_∞ , and hence any quantity depending only on Π_∞ , is determined by a and the sign $\omega_\Pi(-1)$.

3.2.3 Conductors and L -functions

For $N \geq 1$ set

$$\mathcal{U}_1(N) = \left\{ g \in \mathrm{GL}_3(\hat{\mathbb{Z}}) : g \equiv \begin{pmatrix} * & * & * \\ * & * & * \\ 0 & 0 & 1 \end{pmatrix} \pmod{N} \right\}.$$

By the local newvector theory of Jacquet–Piatetski-Shapiro–Shalika there is an N for which $\Pi_f^{\mathcal{U}_1(N)} \neq 0$. If N_Π is the smallest such N then these invariants are one-dimensional, and we call N_Π the *conductor* of Π . We write $N_\Pi^{(p)}$ for its prime-to- p part.

We let $L(\Pi, s) = \prod_{\ell < \infty} L(\Pi_\ell, s)$ denote the standard L -function of Π (without Archimedean factors), in the analytic normalisation, so the Euler product converges for $\Re(s) > 1$. Since Π is both C - and L -algebraic, it is definable over a number field E (its *field of rationality*), and $L(\Pi_\ell, s) = P_\ell(\ell^{-s})^{-1}$ with $P_\ell(X) \in E[X]$ of degree ≤ 3 . For unramified ℓ we write $P_\ell(X) = (1 - \alpha_\ell X)(1 - \beta_\ell X)(1 - \gamma_\ell X)$, where the Satake parameters $\alpha_\ell, \beta_\ell, \gamma_\ell$ are units outside ℓ and have ℓ -adic valuation $\geq -1 - a$. We write $L^{(p)}(\Pi \times \eta, s)$ for the L -function without its Euler factor at p .

3.2.4 Critical values

For a Dirichlet character η , the critical values of $L(\Pi \times \eta, s)$ are at $s = t$ for integers t with either $-a \leq t \leq 0$ and $(-1)^t = \omega_\Pi \eta(-1)$, or $1 \leq t \leq 1 + a$ and $(-1)^t = -\omega_\Pi \eta(-1)$ [LW26, Proposition 2.4]. We thus set

$$\begin{aligned} \mathrm{Crit}^-(\Pi) &= \{(-j, \eta) : 0 \leq j \leq a, (-1)^j = \omega_\Pi \eta(-1)\}, \\ \mathrm{Crit}^+(\Pi) &= \{(j+1, \eta) : 0 \leq j \leq a, (-1)^j = \omega_\Pi \eta(-1)\}, \end{aligned}$$

the critical values in the left and right halves of the critical strip, and write $\mathrm{Crit}_p^\pm(\Pi) \subset \mathrm{Crit}^\pm(\Pi)$ for the subsets where η has p -power conductor.

3.2.5 Rationality and the factor at infinity

Following [LW26, Definition 2.6] (cf. [Coa89, §1]), for $(-j, \eta) \in \mathrm{Crit}^-(\Pi)$ define the modified Euler factor at infinity

$$e_\infty(\Pi_\infty \times \eta_\infty, -j) := i^{j-a-1} \Gamma_{\mathbb{C}}(a+1-j) = 2(2\pi i)^{j-a-1} \Gamma(a+1-j),$$

where $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s} \Gamma(s)$. For $(j+1, \eta) \in \mathrm{Crit}^+(\Pi)$ define

$$e_\infty(\Pi_\infty \times \eta_\infty, j+1) := (-i)^{-j-a-2} \Gamma_{\mathbb{C}}(a+2+j) \cdot \frac{\Gamma_{\mathbb{R}}(j+1+\epsilon)}{(-i)^\epsilon \Gamma_{\mathbb{R}}(\epsilon-j)},$$

where $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2)$ and $\epsilon \in \{0, 1\}$ with $\epsilon \equiv j+1 \pmod{2}$.

Work of Mahnkopf [Mah98, Mah00] and Kasten–Schmidt [KS13] shows that there is a complex period $\Omega_\Pi^- \in \mathbb{C}^\times$ and, for each j , an inexplicit non-zero scalar $\tilde{e}_\infty(\Pi_\infty \times \eta_\infty, -j) \in \mathbb{C}^\times$ such that

$$\tilde{e}_\infty(\Pi_\infty \times \eta_\infty, -j) \frac{L(\Pi \times \eta, -j)}{\Omega_\Pi^-} \in E[\eta, G(\eta)]$$

for all $(-j, \eta) \in \mathrm{Crit}^-(\Pi)$. The main theorem of [LW26] includes the statement that, after a suitable normalisation of ζ_∞ , one has $\tilde{e}_\infty = e_\infty$. We will inherit this in Section 9.

3.3 Hecke operators and P -refinements

Let p be a prime, and recall the maximal parabolics $P_1, P_2 \subset \mathrm{GL}_3$ from Section 2.1. For $i = 1, 2$, let $\mathcal{J}_{p,i}$ denote the parahoric subgroup of $\mathrm{GL}_3(\mathbb{Z}_p)$ associated to P_i . For any open compact $K_p \subset \mathcal{J}_{p,i}$ admitting an Iwahori decomposition with respect to P_i (see Definition 4.5.2) we have normalised Hecke operators

$$U_{p,i} = p^a [K_p \tau_i K_p], \quad \tau_1 = \begin{pmatrix} p & & \\ & 1 & \\ & & 1 \end{pmatrix}, \quad \tau_2 = \begin{pmatrix} p & & \\ & p & \\ & & 1 \end{pmatrix} \in \mathrm{GL}_3(\mathbb{Q}_p),$$

acting on $\Pi_p^{K_p}$. The normalisation factor p^a ensures that these operators preserve integral lattices in Betti cohomology (cf. Section 4.5).

As in [LW26], we denote by $J_{P_i}(\Pi_p)$ the unitarily normalised Jacquet module of Π_p at P_i , a smooth admissible representation of $(\mathrm{GL}_i \times \mathrm{GL}_{3-i})(\mathbb{Q}_p)$.

Definition 3.3.1. For $i = 1, 2$, a P_i -refinement of Π_p is a choice of irreducible representation $\sigma_p \times \sigma'_p$ of $(\mathrm{GL}_i \times \mathrm{GL}_{3-i})(\mathbb{Q}_p)$ appearing as a subrepresentation of $J_{P_i}(\Pi_p)$.

Remark 3.3.2. This is the natural generalisation of an Iwahori refinement of GL_n , which consists of a choice of character of the torus T appearing in $J_B(\Pi_p)$ (in the unramified case, a full ordering of the Satake parameters).

We write $\tilde{\Pi}$ for a pair (Π, σ_p) of Π together with a P_i -refinement; for a given Π_p and i , each of σ_p, σ'_p determines the other, so we usually specify only σ_p , and we say the refinement is *unramified* if σ_p is. Fix an embedding $E \hookrightarrow \mathbb{Q}_p$.

Definition 3.3.3. The *slope* of the P_i -refinement $\sigma_p \times \sigma'_p$ is $v_p(\omega_{\sigma_p}(p))$, where ω_{σ_p} is the central character of σ_p .

For a RACAR of weight $(a, 0, -a)$, the slope of any P_i -refinement lies in the interval $[-1 - a, 1 + a]$. The key assumption of [LW26] was that Π is *nearly ordinary*: that the slope is exactly $-1 - a$, the minimal value. This is the assumption we remove in this article, replacing it by the following.

Definition 3.3.4. We say the P_i -refinement $\sigma_p \times \sigma'_p$ has *small slope* if its slope is < 0 ; equivalently, if

$$v_p(p^{a+1} \omega_{\sigma_p}(p)) < a + 1.$$

As in the GL_2 case, where one usually imposes a p -regularity assumption that the U_p -eigenvalues be distinct, we impose:

Definition 3.3.5. A P_i -refinement $\sigma_p \times \sigma'_p$ is *regular* if all irreducible subquotients of $J_{P_i}(\Pi_p)/(\sigma_p \times \sigma'_p)$ have central characters different from that of $\sigma_p \times \sigma'_p$.

Example 3.3.6. In the style of [LW26, Example 2.13], we briefly recall the classification of generic representations of $\mathrm{GL}_3(\mathbb{Q}_p)$ and describe which of their regular refinements are of small slope.

- If Π_p is supercuspidal, it admits no P_i refinements.
- If $\Pi_p = \mathrm{St}_3 \otimes \lambda$ is a twist of the Steinberg representation by a character λ , it admits a unique P_1 and a unique P_2 refinement, both of which has slope -1 and are thus of small slope for all a .
- If Π_p is parabolically induced from a representation of the form $\theta \times \pi$ of $\mathrm{GL}_1 \times \mathrm{GL}_2$, where θ is a character and π is a supercuspidal representation of GL_2 , then its unique P_1 refinement is θ and its unique P_2 refinement is π both of which are regular. Their slopes are $v_p(\theta(p))$ and $-v_p(\theta(p))$ respectively, so the former is of small slope when $v_p(\theta(p)) < 0$ and the later when $v_p(\theta(p)) > 0$.
- If Π_p is irreducibly induced from $\theta \times (\mathrm{St}_2 \otimes \lambda)$, then it has two P_1 -refinements, θ and $\lambda \cdot |\cdot|^{1/2}$, of slopes $v_p(\theta(p))$ and $-\frac{1}{2}(1 + v_p(\theta(p)))$, and two P_2 -refinements, $\mathrm{St}_2 \otimes \lambda$ and $\mathrm{Ind}(\theta \otimes \lambda \cdot |\cdot|^{1/2})$, of slopes $-v_p(\theta(p))$ and $1/2(v_p(\theta(p)) - 1)$, all four of which are regular. Thus, it has at least one P_1 and one P_2 refinement of small slope.

- If Π_p is an irreducible principal series representation, induced from a character $\chi_1 \times \chi_2 \times \chi_3$, then its P_1 -refinements are the χ_i , of slope $v_p(\chi_i(p))$, and its P_2 -refinements are the $\text{Ind}(\chi_j \otimes \chi_k)$, of slope $-v_p(\chi_i(p))$, where $\{i, j, k\} = \{1, 2, 3\}$. The regular ones are the ones for which $\chi_i \neq \chi_j, \chi_k$. Π_p has a small slope P_1 refinement and a small slope P_2 refinement unless $v_p(\chi_i(p)) = 0$ for all i .

Remark 3.3.7. Small slope refinements need not exist. For example, if Π_p is unramified with Satake parameters $\alpha_p, \beta_p, \gamma_p$, the P_1 -refinements are the unramified characters sending p to one of these parameters. Since $v_p(\alpha_p) + v_p(\beta_p) + v_p(\gamma_p) = v_p(\omega_\Pi(p)) = 0$ it can happen that all three valuations vanish, in which case every P_1 -refinement has slope 0, i.e. normalised slope exactly $a + 1$.

Remark 3.3.8. If, for example, $\Pi_p = \text{Ind}(\chi_1 \otimes \chi_2 \otimes \chi_3)$ is an unramified principal series and we consider the P_1 -refinement χ_j , regularity amounts to $\chi_j \neq \chi_k$ for $k \neq j$. Regularity is automatic for nearly ordinary refinements; at finite slope we impose it throughout, so that the relevant Hecke eigenspaces are genuine (not merely generalised) eigenspaces.

3.3.1 P_1 -refined newvectors

From now on fix a regular unramified P_1 -refinement σ_p , determined by $\alpha_p := \sigma_p(p) \neq 0$. (This is no loss of generality: as in [LW26, Proposition 2.19], since σ_p is a character of $\mathbb{Q}_p^\times$ we may twist Π by a Dirichlet character to make it unramified, and the conjectural interpolation property is invariant under such twists.) For $r \geq 0$ consider the group

$$\mathcal{U}_{1,p}^{(P_1)}(p^r) = \left\{ g \in \text{GL}_3(\mathbb{Z}_p) : g \equiv 1 \pmod{\begin{pmatrix} * & * & * \\ p & * & * \\ p^R & p^r & p^r \end{pmatrix}} \right\} \subset \text{GL}_3(\mathbb{Z}_p), \quad R = \max\{r, 1\}.$$

This group admits an Iwahori decomposition with respect to P_1 , and its intersection with the GL_2 -factor of L_1 is the level- p^r mirabolic subgroup of Casselman's local newvector theory [Cas73]. Combining Casselman's canonical lifting theorem (see [Eme06] for an account) with the GL_2 newvector theorem, one obtains (cf. [LW26, Proposition 2.24]):

Proposition 3.3.9. *For all sufficiently large r there exists $\varphi_p \in \Pi_p^{\mathcal{U}_{1,p}^{(P_1)}(p^r)}$ satisfying $U_{p,1}\varphi_p = p^{1+a}\alpha_p\varphi_p$. The minimal such r , denoted $r(\tilde{\Pi})$, equals the conductor of the $\text{GL}_2(\mathbb{Q}_p)$ -representation σ'_p , and for $r = r(\tilde{\Pi})$ the space of such vectors is one-dimensional.*

We define the P_1 -refined Whittaker newvector W^α to be the unique basis vector of this one-dimensional space in the Whittaker model of Π_p normalised so that $W^\alpha(1) = 1$. Its values along the torus are given in terms of the GL_2 -newvector of σ'_p by [LW26, Proposition 2.25].

3.4 The Coates–Perrin–Riou factor at p

Let $(t, \text{id}) \in \text{Crit}^\pm(\Pi)$ and let $i = 1$ for the sign $-$ and $i = 2$ for the sign $+$. Write $\psi^- = \psi$ and $\psi^+ = \psi^{-1}$. For a P_i -refinement σ_p of Π_p , following [Coa89, §2] and [LW26, Definition 2.16] we define the modified Euler factor at p :

$$e_p(\Pi_p, \sigma_p, t) := \gamma_p(\sigma_p, \psi^\pm, t)^{-1} = \frac{L(\sigma_p, t)}{L(\sigma_p^\vee, 1-t) \varepsilon_p(\sigma_p, \psi^\pm, t)},$$

where γ_p is the local γ -factor, and the sign of the additive character matches the sign of the critical region. Since our refinements are fixed throughout, we drop σ_p (and $\pm$) from the notation and write $e_p(\Pi_p, t)$, and for twists by characters η_p of $\text{GL}_3(\mathbb{Q}_p)$ we write $e_p(\Pi_p \times \eta_p, t) := e_p(\Pi_p \times \eta_p, \sigma_p \times \eta_p, t)$.

Example 3.4.1. If σ_p is an unramified P_1 -refinement with $\sigma_p(p) = \alpha_p \neq 0$, then for $(-j, \eta) \in \text{Crit}_p^-(\Pi)$,

$$e_p(\Pi_p \times \eta_p, -j) = \begin{cases} G(\eta^{-1}) \cdot (p^{j+1}\alpha_p)^{-n} & \text{cond}(\eta) = p^n > 1, \\ \frac{1 - p^{-j-1}\alpha_p^{-1}}{1 - p^j\alpha_p} & \eta = 1. \end{cases}$$

4 Locally symmetric spaces and cohomology

We define the locally symmetric spaces, open compact subgroups and local systems that we will use in this article.

4.1 Locally symmetric spaces

For any reductive group A over $\mathbb{Q}$ and any open compact subgroup $K \subset A(\mathbb{A}_f)$, define

$$Y^A(K) := A^+(\mathbb{Q}) \backslash ((A(\mathbb{A}_f)/K) \times \mathcal{H}_A^+),$$

where a superscript $(\cdot)^+$ indicates the connected component of the identity in the real topology, $\mathcal{H}_A^+ = A^+(\mathbb{R})/K_{A,\infty}^\circ Z_A^+(\mathbb{R})$, $K_{A,\infty} \subset A(\mathbb{R})$ is a maximal compact subgroup, and Z_A is the centre of A .

Since $\iota(Z_H) \not\subset Z_G$, the embedding ι does not induce a map $\mathcal{H}_H^+ \rightarrow \mathcal{H}_G^+$ of symmetric spaces. To account for this, as in [LW26], we consider the auxiliary space $\tilde{\mathcal{H}}_H^+ := H^+(\mathbb{R})/K_{H,\infty}^\circ \iota^{-1}(Z_G^+(\mathbb{R}))$, which, since $\iota^{-1}(Z_G) \subset Z_H$ admits natural maps

$$\begin{array}{ccc} & \tilde{\mathcal{H}}_H^+ & \\ f \swarrow & & \searrow \iota \\ \mathcal{H}_H^+ & & \mathcal{H}_G^+, \end{array} \tag{7}$$

the left map being the natural quotient and the right map induced by ι . For any open compact $K_H \subset H(\mathbb{A}_f)$ we write

$$\tilde{Y}^H(K_H) = H^+(\mathbb{Q}) \backslash ((H(\mathbb{A}_f)/K_H) \times \tilde{\mathcal{H}}_H^+),$$

and for open compact $K_G \subset G(\mathbb{A}_f)$ we obtain a diagram

$$\begin{array}{ccc} & \tilde{Y}^H(K_G \cap H) & \\ f \swarrow & & \searrow \iota \\ Y^H(K_G \cap H) & & Y^G(K_G). \end{array}$$

The spaces Y^H (resp. $\tilde{Y}^H$, resp. Y^G) have dimension 2 (resp. 3, resp. 5) as real manifolds; if K_G is sufficiently small the map ι on the right is a closed embedding of codimension 2 (cf. [LW26, Theorem 6.10] and [Loe21, §4.6]).

4.2 Connected components

Definition 4.2.1. Let $A \in \{G, H, \mathrm{GL}_2\}$ and let $K \subset A(\mathbb{A}_f)$ be a compact subgroup. For integers $M_1, M_2 \geq 1$ define

$$K(M_1, M_2) = \{k \in K : \nu_i(k) \in 1 + M_i \hat{\mathbb{Z}}, i = 1, 2\}.$$

When $M_2 = 1$, we write $K(M_1, 1) = K(M_1)$.

Let $K \subset A(\mathbb{A}_f)$ be open compact, $M_1, M_2 \geq 1$ and suppose the map

$$\begin{aligned} K &\rightarrow (\mathbb{Z}/M_1\mathbb{Z})^\times \times (\mathbb{Z}/M_2\mathbb{Z})^\times \\ k &\mapsto (\nu_1(k), \nu_2(k)) \end{aligned}$$

is surjective. We regard ν_i as a map

$$\nu_i : A^+(\mathbb{Q}) \backslash A(\mathbb{A}_f)/K(M_1, M_2) \rightarrow \mathbb{Q}_{>0}^\times \backslash \mathbb{A}_f^\times / ((1 + M_i \hat{\mathbb{Z}}) \prod_{\ell \nmid M_i} \mathbb{Z}_\ell^\times) \cong (\mathbb{Z}/M_i\mathbb{Z})^\times,$$

the last identification being normalised so that a uniformiser ϖ_ℓ at $\ell \nmid M_i$ maps to $\ell \bmod M_i$. The following lemma is a special case of a well-known general phenomenon.

Lemma 4.2.2. *The map*

$$\Upsilon = (\nu_1, \nu_2, \text{pr}) : Y^A(K(M_1, M_2)) \xrightarrow{\sim} (\mathbb{Z}/M_1\mathbb{Z})^\times \times (\mathbb{Z}/M_2\mathbb{Z})^\times \times Y^A(K)$$

is an isomorphism, equivariant for the natural action of K on the source and the action

$$k \cdot (x_1, x_2, y) = (\nu_1(k)x_1, \nu_2(k)x_2, y)$$

on the target.

4.3 Open compact subgroups

We now fix the level structures used in the construction. Fix $n \geq 1$ and recall $r = r(\tilde{\Pi})$ from Proposition 3.3.9.

4.3.1 Subgroups of GL_2

Let $\mathcal{U}^{\text{GL}_2, (p)} \subset \text{GL}_2(\mathbb{A}_f^{(p)})$ be open compact and define

$$\mathcal{U}_n^{\text{GL}_2} := \{A \in \text{GL}_2(\mathbb{Z}_p) : A \equiv \begin{pmatrix} * & * \\ * & 1 \end{pmatrix} \pmod{p^n}\} \times \mathcal{U}^{\text{GL}_2, (p)} \subset \text{GL}_2(\mathbb{A}_f).$$

4.3.2 Subgroups of H

Let $\mathcal{U}^{H, (p)} \subset H(\mathbb{A}_f^{(p)})$ be open-compact and define for $t, n \geq 1$ the following subgroups of $H(\mathbb{A}_f)$:

$$\begin{aligned} Q_{H, t, n}^0 &:= \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \times (z) : b \equiv 0 \pmod{p^n}, c \equiv 0 \pmod{p^t}, d \equiv 1 \pmod{p^t} \right\} \times \mathcal{U}^{H, (p)}, \\ Q_{H, \infty, n}^0 &:= \bigcap_t Q_{H, t, n}^0, \\ \mathcal{U}_1^H(p^t) &:= \{A \in H(\mathbb{Z}_p) : A \equiv \begin{pmatrix} * & * \\ * & 1 \end{pmatrix} \times (*) \pmod{p^t}\} \times \mathcal{U}^{H, (p)}. \end{aligned}$$

We use the notation of Definition 4.2.1 for these groups: for example $Q_{H, t, n}^0(p^n) = \{h \in Q_{H, t, n}^0 : \nu_1(h) \equiv 1 \pmod{p^n}\}$.

4.3.3 Subgroups of GL_3 and G

Let $\mathcal{U}^{\text{GL}_3, (p)} \subset \text{GL}_3(\mathbb{A}_f^{(p)})$ be open compact and define the following subgroups of $\text{GL}_3(\mathbb{A}_f)$:

$$\begin{aligned} \mathcal{U} &:= \mathcal{U}_{1, p}^{(P_1)}(p^r) \times \mathcal{U}^{\text{GL}_3, (p)} = \left\{ g \in \text{GL}_3(\mathbb{Z}_p) : g \equiv 1 \pmod{\begin{pmatrix} * & * & * \\ p^R & p^r & p^r \end{pmatrix}} \right\} \times \mathcal{U}^{\text{GL}_3, (p)}, \\ \mathcal{U}_n^{\text{GL}_3} &:= \left\{ g \in \text{GL}_3(\mathbb{Z}_p) : g \equiv 1 \pmod{\begin{pmatrix} * & p^n & p^n \\ * & p^r & p^r \end{pmatrix}} \right\} \times \mathcal{U}^{\text{GL}_3, (p)}, \\ \mathcal{V}_n^{\text{GL}_3} &:= \tau^{-n} \mathcal{U}_n^{\text{GL}_3} \tau^n = \left\{ g \equiv 1 \pmod{\begin{pmatrix} * & * & * \\ p^n & p^r & p^r \\ p^{n+r} & p^r & p^r \end{pmatrix}} \right\} \times \mathcal{U}^{\text{GL}_3, (p)}, \end{aligned}$$

where $R = \max\{r, 1\}$. Note that $\mathcal{V}_n^{\text{GL}_3} \subset \mathcal{U}$ for all $n \geq 1$, and that $\mathcal{U}_{n+1}^{\text{GL}_3} = \mathcal{U}_n^{\text{GL}_3} \cap \tau \mathcal{U}_n^{\text{GL}_3} \tau^{-1}$. We define

$$\begin{aligned} \mathcal{U}_n^G &= \mathcal{U}_n^{\text{GL}_3} \times \hat{\mathbb{Z}}^\times \\ \mathcal{V}_n^G &= \tau^{-n} \mathcal{U}_n^G \tau^n \\ \mathcal{U}^{G, (p)} &= \mathcal{U}^{\text{GL}_3, (p)} \times \prod_{\ell \neq p} \mathbb{Z}_\ell^\times. \end{aligned}$$

The following is the analogue of [LW26, Proposition 6.3].

Lemma 4.3.1. *Let $t \geq \max(n, r)$ and $h = \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}, z\right) \in H(\mathbb{Z}_p)$. Then*

1. $\mathcal{U}_1^H(p^t) \cap u\mathcal{U}_n^G u^{-1}$ consists of those $h \in \mathcal{U}_1^H(p^t)$ with $b \equiv 0$ and $a \equiv z \pmod{p^n}$. Its index in $\mathcal{U}_1^H(p^t)$ is $p^{2n-1}(p-1)$,
2. We have $Q_{H,t,n}^0(p^n) = \mathcal{U}_1^H(p^t) \cap u\mathcal{U}_n^G u^{-1}$. In particular, if we have $\nu_2(\mathcal{U}^{H,(p)}) \in 1 + D\hat{\mathbb{Z}}$ for some prime-to- p D , then

$$\mathcal{U}_1^H(p^t) \cap u\mathcal{U}_n^G u^{-1} \subset u\mathcal{U}_n^G(p^n, D)u^{-1}.$$

Proof. Part (1) is [LW26, Proposition 6.3], the rest is easily checked. $\square$

4.4 Local systems on locally symmetric spaces

Let A, K be as above and let V be a $\mathbb{Q}$ -linear algebraic representation of A . There are three standard constructions of local systems on $Y^A(K)$ attached to V :

- a local system $\mathcal{V}_{\mathbb{Q}}$ of $\mathbb{Q}$ -vector spaces, defined as the locally constant sections of

$$A^+(\mathbb{Q}) \backslash [(A(\mathbb{A}_f) \times \mathcal{H}_A^+) \times V(\mathbb{Q})] / K \rightarrow Y^A(K),$$

with action $\gamma \cdot [(g, z), v] \cdot k = [(\gamma g k, \gamma z), \gamma v]$,

- a local system $\mathcal{V}_{\mathbb{R}}$ of $\mathbb{R}$ -vector spaces, defined as the locally constant sections of

$$A^+(\mathbb{Q}) \backslash [(A(\mathbb{A}_f) \times A^+(\mathbb{R})) \times V(\mathbb{R})] / K K_{A,\infty}^{\circ} Z_{A,\infty}^{\circ} \rightarrow Y^A(K),$$

with action $\gamma \cdot [(g, z), v] \cdot k k_{\infty} z_{\infty} = [(\gamma g k, \gamma z k_{\infty} z_{\infty}), (k_{\infty} z_{\infty})^{-1} v]$,

- a local system $\mathcal{V}_{\mathbb{Q}_p}$ of $\mathbb{Q}_p$ -vector spaces, defined as the locally constant sections of

$$A^+(\mathbb{Q}) \backslash [(A(\mathbb{A}_f) \times \mathcal{H}_A^+) \times V(\mathbb{Q}_p)] / K \rightarrow Y^A(K),$$

with action $\gamma \cdot [(g, z), v] \cdot k = [(\gamma g k, \gamma z), k_p^{-1} v]$, where k_p is the p -component of k .

If V carries a K_p -stable lattice $V_{\mathbb{Z}_p}$ (e.g. an admissible lattice as in Section 2.3), the third construction yields a local system $\mathcal{V}_{\mathbb{Z}_p}$ of $\mathbb{Z}_p$ -modules, and similarly with coefficients in $\mathcal{O}_L$ for $L/\mathbb{Q}_p$ finite. There are comparison isomorphisms

$$\mathcal{V}_{\mathbb{Q}} \otimes \mathbb{Q}_p \cong \mathcal{V}_{\mathbb{Q}_p}, \quad \mathcal{V}_{\mathbb{Q}} \otimes \mathbb{R} \cong \mathcal{V}_{\mathbb{R}}, \quad (8)$$

the first given on sections by $[(g, z), v] \mapsto [(g, z), g_p^{-1} v]$, see [Urb11, §1.2].

Lemma 4.4.1. *There is a natural injective map*

$$V^{K_p} \rightarrow H^0(Y^A(K), \mathcal{V}_{\mathbb{Q}_p})$$

sending $v \mapsto \varphi_v$, where

$$\varphi_v([g, z]) = ([g, z], v).$$

Moreover, if $t \in A(\mathbb{Q}_p)$ normalises K then the above map is equivariant for the action of t on both sides, with the natural pullback action on $\varphi \in H^0(Y^A(K), \mathcal{V}_{\mathbb{Q}_p})$, given explicitly by

$$(t \cdot \varphi)([g, z]) = \varphi([gt, z]) \cdot t^{-1}.$$

Proof. The map is well-defined since for $\gamma \in A(\mathbb{Q}), k \in K$ we have $\varphi_v([\gamma g k, \gamma z]) = ([\gamma g k, \gamma z], v) = ([g, z], k_p v) = ([g, z], v)$ since $v \in V^{K_p}$ and is clearly injective. Given $t \in A(\mathbb{Q}_p)$ normalising K we have

$$\begin{aligned} \varphi_{t \cdot v}([g, z]) &= ([g, z], t \cdot v) \\ &= ([gt t^{-1}, z], t \cdot v) \\ &= ([gt, z], v) \cdot t^{-1} \\ &= \varphi_v([gt, z]) \cdot t^{-1}. \end{aligned}$$

$\square$

Remark 4.4.2. More generally for $g \in A(\mathbb{Q}_p)$ the above map is compatible with the natural maps $V^{K_p} \rightarrow V^{g^{-1}K_p g}$ and $H^0(Y^A(K), \mathcal{V}_{\mathbb{Q}_p}) \rightarrow H^0(Y^A(g^{-1}Kg), \mathcal{V}_{\mathbb{Q}_p})$.

Definition 4.4.3. We define two families of sections.

1. For $n \geq 1$, let $K_n \subset G(\mathbb{A}_f)$ be an open compact subgroup whose reduction mod p^n is contained in $Q_H^0(\mathbb{Z}/p^n\mathbb{Z})$. For $0 \leq j \leq a$, we write $f_n^{a,j}$ for the image of $f^{a,j}$ in $(V_{\lambda, \mathbb{Z}_p} \otimes \nu_1^{-j})/p^n$ and

$$\varphi_n^{a,j} := \varphi_{f_n^{a,j}} \in H^0(Y^G(K_n), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)/p^n).$$

2. Let $K_n \subset G(\mathbb{A}_f)$ that reduces mod p^n to a subgroup of the kernel of ν_1^j , for some $j \geq 0$. We write

$$\varphi_{j,n} := \varphi_1 \in H^0(Y^G(K_n), \nu_1^j/p^n),$$

that is, the image of the element $1 \in \mathbb{Z}_p \otimes \nu_1^j$ under the map of Lemma 4.4.1.

4.5 Cohomology, translation maps and Hecke operators

The constructions above are compatible with the projections $Y^A(K') \rightarrow Y^A(K)$ for $K' \subset K$, inducing pullback maps

$$H^i(Y^A(K), \mathcal{V}_?) \rightarrow H^i(Y^A(K'), \mathcal{V}_?), \quad ? \in \{\mathbb{Q}, \mathbb{Q}_p, \mathbb{R}\}. \quad (9)$$

For $g \in A(\mathbb{A}_f)$ with $g^{-1}Kg \subset K'$, right translation gives maps

$$\begin{aligned} [g]_{K'}^K : Y^A(K) &\rightarrow Y^A(K'), \\ [h, z] &\mapsto [hg, z], \end{aligned}$$

and hence pullback and pushforward maps

$$\begin{aligned} ([g]_{K'}^K)^* : H^i(Y^A(K'), \mathcal{V}_?) &\rightarrow H^i(Y^A(K), \mathcal{V}_?), \\ ([g]_{K'}^K)_* : H^i(Y^A(K), \mathcal{V}_?) &\rightarrow H^i(Y^A(K'), \mathcal{V}_?), \end{aligned}$$

We write $[g]$ for $[g]_{K'}^K$, when $g^{-1}Kg = K'$.

Remark 4.5.1. The maps $([1]_{K'}^K)^*$ and $([1]_{K'}^K)_*$ are the usual restriction and norm maps.

The pullback maps assemble into an action of $A(\mathbb{A}_f)$ on $H^i(Y^A, \mathcal{V}_?) := \varinjlim_K H^i(Y^A(K), \mathcal{V}_?)$, and the cohomology at each level carries an action of the Hecke algebra $\mathbb{Q}[K \backslash A(\mathbb{A}_f)/K]$, compatibly with (8).

Definition 4.5.2. An open compact $K_p \subset \mathrm{GL}_3(\mathbb{Z}_p)$ admits an Iwahori decomposition with respect to P_1 if multiplication induces a bijection

$$(\bar{N}_1(\mathbb{Z}_p) \cap K_p) \times (L_1(\mathbb{Z}_p) \cap K_p) \times (N_1(\mathbb{Z}_p) \cap K_p) \xrightarrow{\sim} K_p.$$

Associated to $u\tau^n \in G(\mathbb{Q}_p)$ we have a pushforward map $[u\tau^n]_* : H^3(Y^G(K), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Q}_p)) \rightarrow H^3(Y^G((u\tau^n)^{-1}Ku\tau^n), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Q}_p))$. We can normalise this map to preserve the integral cohomology using the normalised $*$ -action of τ^{-n} as in Section 2.4.2.

Definition 4.5.3. Write

$$[u\tau^n]_{*,j} := p^{(a-j)n} [u\tau^n]_* : H^3(Y^G(K), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \rightarrow H^3(Y^G((u\tau^n)^{-1}Ku\tau^n), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)).$$

Now take $K = K^{(p)}K_p \subset \mathrm{GL}_3(\mathbb{A}_f)$ with K_p admitting an Iwahori decomposition with respect to P_1 . We have the correspondence

$$\begin{array}{ccc} & Y^{\mathrm{GL}_3}(K \cap \tau K \tau^{-1}) & \\ \swarrow [1] & & \searrow [\tau] \\ Y^{\mathrm{GL}_3}(K) & & Y^{\mathrm{GL}_3}(K), \end{array}$$

where the left map is the natural projection and the right map is translation by τ followed by the projection from level $\tau^{-1}K\tau \cap K$. We define Hecke operators,

$$U'_p := p^{a-j} \left([1]_K^{\tau^{-1}K\tau \cap K} \right)_* \circ [\tau]_* \circ \left([1]_K^{K \cap \tau K \tau^{-1}} \right)^*,$$

on $H^i_\gamma(Y^{\mathrm{GL}_3}(K), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p))$, $? \in \{\emptyset, c\}$, and

$$U_p := p^{a-j} \left([1]_K^{K \cap \tau K \tau^{-1}} \right)_* \circ [\tau^{-1}]_* \circ \left([1]_K^{\tau^{-1}K\tau \cap K} \right)^*,$$

on $H^i_\gamma(Y^{\mathrm{GL}_3}(K), (\mathcal{V}_\lambda \otimes \nu_1^{-j})^\vee(\mathbb{Z}_p))$, $? \in \{\emptyset, c\}$ where the action of $\tau^{\mp 1}$ on the coefficients $\mathcal{V}_{\lambda, \mathbb{Z}_p}$ is the normalised $*$ -action of (5), which preserves the integral lattice. With this normalisation $U_p = U_{p,1}$ agrees with the operator of Section 3.3 on the Π -isotypic part. Poincaré duality gives a perfect pairing

$$\langle -, - \rangle_K : \frac{H^3(Y^{\mathrm{GL}_3}(K), \mathcal{V}_\lambda(R))}{(\text{torsion})} \times \frac{H^2_c(Y^{\mathrm{GL}_3}(K), \mathcal{V}_\lambda^\vee(R))}{(\text{torsion})} \rightarrow R \quad (10)$$

for any integral extension $R/\mathbb{Z}_p$, given by cup product, the pairing $\mathcal{V}_\lambda \otimes \mathcal{V}_\lambda^\vee \rightarrow R$, and integration over the 5-dimensional manifold $Y^{\mathrm{GL}_3}(K)$. Since pullback and pushforward are adjoint under Poincaré duality, the operators U_p and U'_p (acting on the two factors) are adjoint to one another under $\langle -, - \rangle_K$.

4.6 Iwasawa cohomology

We extend cohomology to compact subgroups which are not necessarily open.

Definition 4.6.1. Let V be a $\mathbb{Q}_p$ -linear algebraic representation of A , let $V_{\mathbb{Z}_p} \subset V$ be an admissible lattice, and let $\mathcal{V}_{\mathbb{Z}_p}$ be the associated local system. Let $B = U^{(p)}B_p \subset A(\mathbb{A}_f)$, where $U^{(p)} \subset A(\mathbb{A}_f^{(p)})$ is open compact and $B_p \subset A(\mathbb{Z}_p)$ is compact. The *Iwasawa cohomology* of B is

$$H^i_{\mathrm{Iw}}(Y^A(B), \mathcal{V}_{\mathbb{Z}_p}) := \varprojlim_{U_p \supset B_p} H^i(Y^A(U^{(p)}U_p), \mathcal{V}_{\mathbb{Z}_p}),$$

the limit over (a cofinal system of) open compact $U_p \subset A(\mathbb{Z}_p)$ containing B_p , with respect to the norm maps. We define $H^i_{\mathrm{Iw}}(\tilde{Y}^H(B), -)$ for $B \subset H(\mathbb{A}_f)$ in the same way.

Let B and $\mathcal{V}_{\mathbb{Z}_p}$ be as in the definition, and for $n \geq 1$ let $U_n \subset A(\mathbb{Z}_p)$ be the preimage of the image of B_p in $A(\mathbb{Z}/p^n\mathbb{Z})$. Let $W_{\mathbb{Z}_p}$ be an admissible lattice in another $\mathbb{Q}_p$ -linear algebraic representation of A . For $w \in W_{\mathbb{Z}_p}^{B_p}$, let w_n be its image in $(W_{\mathbb{Z}_p}/p^n)^{U_n}$, and let $\varphi_{w_n} \in H^0(Y^A(U^{(p)}U_n), \mathcal{W}_{\mathbb{Z}_p}/p^n)$ be the section attached to w_n by the mod p^n analogue of Lemma 4.4.1. Then there is a cup product pairing

$$H^i_{\mathrm{Iw}}(Y^A(B), \mathcal{V}_{\mathbb{Z}_p}) \times W_{\mathbb{Z}_p}^{B_p} \rightarrow H^i_{\mathrm{Iw}}(Y^A(B), \mathcal{V}_{\mathbb{Z}_p} \otimes \mathcal{W}_{\mathbb{Z}_p}), \quad (z, w) \mapsto \varprojlim_n (z U_n \bmod p^n) \cup \varphi_{w_n}. \quad (11)$$

4.7 Twist operators

Let $K \subset G(\mathbb{A}_f)$ be an open compact subgroup satisfying $\nu_1(K_p) = 1 + p^n\mathbb{Z}_p$ for some $n \geq 1$ and fix a set representatives $\{g_1, \dots, g_d\}$ for the map

$$\nu_1 \times \nu_2 : G^+(\mathbb{Q}) \backslash G(\mathbb{A}_f) \times \mathcal{H}_G^+/K \rightarrow (\mathbb{Q}_{>0}^\times)^2 \backslash (\mathbb{A}_f^\times)^2 / (\nu_1 \times \nu_2)(K) =: \Delta.$$

Definition 4.7.1. For $j \in \mathbb{Z}$ define the *twisting map*

$$\begin{aligned} \text{tw}_j : \text{Maps}(\Delta, \mathbb{Z}_p) &\rightarrow \text{Maps}(\Delta, \mathbb{Z}_p), \\ f &\mapsto \left(g_i \mapsto (\|\nu_1(g_i)\| \nu_1(g_{i,p}))^{-j} f((\nu_1 \times \nu_2)(g_i)) \right). \end{aligned}$$

Note that tw_j depends on the choice of g_i ; replacing g_i with $\tilde{g}_i$ in the same equivalence class changes $(\|\nu_1(g_i)\| \nu_1(g_{i,p}))^{-j}$ by an element of $1 + p^n \mathbb{Z}_p$. Now suppose K surjects onto $(\mathbb{Z}/p^n \mathbb{Z})^\times \times (\mathbb{Z}/D\mathbb{Z})^\times$ via $\nu_1 \times \nu_2$, with $D \geq 1$ prime to p , and consider the level $K(p^n, D)$. By Lemma 4.2.2,

$$Y^G(K(p^n, D)) \cong \Delta_{1,n} \times \Delta_2 \times Y^G(K), \quad \Delta_{1,n} \cong (\mathbb{Z}/p^n \mathbb{Z})^\times, \Delta_2 := (\mathbb{Z}/D\mathbb{Z})^\times,$$

where the identifications on the right send uniformisers ϖ_ℓ for ℓ away from p (resp. D) to $\ell \bmod p^n$ (resp. $\bmod D$). For each $(x, y) \in \Delta_{1,n} \times \Delta_2$ fix a base point $g_{x,y} \in G(\mathbb{A}_f)$ in the corresponding union of connected components.

Lemma 4.7.2. *There is a canonical isomorphism*

$$\Psi_j : H^3(Y^G(K(p^n, D)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \xrightarrow{\sim} H^3(Y^G(K), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \otimes \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p). \quad (12)$$

Proof. We sketch the proof. Write $Y' = Y^G(K(p^n, D))$ and $Y = Y^G(K)$, let $\pi : Y' \rightarrow Y$ for the natural finite covering map and write $\mathbb{Z}_p$ for the locally constant sheaf associated to the trivial representation. By the projection formula and the Leray spectral sequence we have

$$H^i(Y, (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p) \otimes \pi_* \mathbb{Z}_p) \cong H^i(Y', (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)).$$

We have a canonical (independent of the choice of $g_{x,y}$) isomorphism of constant sheaves $\pi_* \mathbb{Z}_p \cong \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p)$ and thus an isomorphism

$$\Psi_j : H^i(Y, (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p) \otimes \pi_* \mathbb{Z}_p) \cong H^i(Y, (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \otimes_{\mathbb{Z}_p} \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p).$$

□

The choice of $\{g_{x,y} : (x, y) \in \Delta_{1,n} \times \Delta_2\}$ fixes a twisting map

$$\text{tw}_j : \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p) \rightarrow \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p)$$

as in Definition 4.7.1.

Lemma 4.7.3. *Let $j \in \mathbb{Z}$, $A \in \{G, H\}$ and $\mathcal{V}$ the locally constant sheaf associated to an algebraic representation of $A(\mathbb{Q}_p)$. For any open compact $K \subset A(\mathbb{A}_f)$ define a section*

$$\mathbb{S}_j^A(g, z) = ([g, z], (\|\nu_1(g)\| \nu_1(g_p))^{-j}) \in H^0(Y^A(K), \nu_1^j)$$

of the locally constant sheaf associated to ν_1^j . Let $K \subset A(\mathbb{A}_f)$, then cup product with $\mathbb{S}_j^A$ defines a map

$$T_j^A := (\cdot \cup \mathbb{S}_j^A) : H^i(Y^A(K), (\mathcal{V} \otimes \nu_1^{-j})(\mathbb{Z}_p)) \rightarrow H^i(Y^A(K), \mathcal{V}(\mathbb{Z}_p)).$$

Then

1. *We have*

$$\iota^* \circ T_j^G = T_j^H \circ \iota^*.$$

We will therefore henceforth write $\mathbb{S}_j$ for $\mathbb{S}_j^A$ and T_j for cup product with this element.

2. Let $K \subset G(\mathbb{A}_f)$, let $\mathbb{S}_{j,n} \in H^0(Y^G(K), \nu_1^j/p^n)$ be the reduction of $\mathbb{S}_j$ modulo p^n and $T_{j,n}$ the cup product with this element, and suppose K surjects onto $(\mathbb{Z}/p^n\mathbb{Z})^\times \times (\mathbb{Z}/D\mathbb{Z})^\times$ via $\nu_1 \times \nu_2$. Writing $M_n = \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p)$, the diagram

$$\begin{array}{ccc} H^3(Y^G(K(p^n, D)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)/p^n) & \xrightarrow{\cup \varphi_{j,n}} & H^3(Y^G(K(p^n, D)), \mathcal{V}_\lambda(\mathbb{Z}_p)/p^n) \\ \Psi_j \downarrow & & \downarrow \Psi_0 \\ H^3(Y^G(K), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)/p^n) \otimes M_n & \xrightarrow{T_{j,n} \otimes \text{tw}_{-j}} & H^3(Y^G(K), \mathcal{V}_\lambda(\mathbb{Z}_p)/p^n) \otimes M_n \end{array}$$

commutes.

3. We have the relation

$$T_j \circ [u\tau^n]_{*,j} = [u\tau^n]_{*,0} \circ T_j.$$

In particular, $U'_p \circ T_j = T_j \circ U'_p$.

Proof. Clear by construction. $\square$

Note that the diagram in part (2) is canonical since the ambiguity in the choices defining tw_j are 1 mod p^n . Indeed, tw_j is given explicitly modulo p^n by multiplication by the function

$$[(x, y)] \mapsto x^j,$$

where we identify $\Delta_{1,n} \times \Delta_2 \cong (\mathbb{Z}/p^n\mathbb{Z})^\times \times (\mathbb{Z}/D\mathbb{Z})^\times$ via $(\delta_1, \delta_2) \mapsto (x, y) = \left((\|\delta_1\| \delta_{1,p})^{-1}, (\|\delta_2\| \delta_{2,D})^{-1} \right) \bmod \left((1 + p^n \hat{\mathbb{Z}}) \times (1 + D\hat{\mathbb{Z}}) \right)$, with $(\delta_1, \delta_2) \in \left(\mathbb{A}_f^\times \right)^2$ a lift of an element of $\Delta_{1,n} \times \Delta_2$.

4.8 Automorphic cohomology classes

Let Π be a RACAR of $\text{GL}_3(\mathbb{A})$ of weight $\lambda = (a, 0, -a)$, and let $K \subset \text{GL}_3(\mathbb{A}_f)$ be open compact with $\Pi_f^K \neq 0$. Via cuspidal cohomology and the choice of generator ζ_∞ of (6) there is a Hecke-equivariant injection (see [LW26, (3.3)])

$$\phi_\Pi^K : \mathcal{W}(\Pi_f)^K \hookrightarrow H_c^2(Y^{\text{GL}_3}(K), \mathcal{V}_{\lambda, \mathbb{Q}}^\vee(\mathbb{C})),$$

compatible with change of K , assembling to a $\text{GL}_3(\mathbb{A}_f)$ -equivariant map

$$\phi_\Pi : \mathcal{W}(\Pi_f) \hookrightarrow H_c^2(Y^{\text{GL}_3}, \mathcal{V}_{\lambda, \mathbb{Q}}^\vee(\mathbb{C}));$$

we write $H_c^2(\Pi, \mathbb{C})$ for its image.

4.8.1 Periods and rationality

There is a minimal number field E , the field of definition of Π , such that

$$H_c^2(\Pi, E) := H_c^2(\Pi, \mathbb{C}) \cap H_c^2(Y^{\text{GL}_3}, \mathcal{V}_{\lambda, \mathbb{Q}}^\vee(E))$$

is non-zero and gives an E -structure on $H_c^2(\Pi, \mathbb{C})$; by strong multiplicity one, E may be taken to be the field generated by the Hecke eigenvalues of Π at unramified primes. Comparing with the Whittaker E -structure defines, by [Clo90, Proposition 3.1], a period $\Theta_\Pi \in \mathbb{C}^\times$ with

$$\phi_\Pi(\mathcal{W}(\Pi_f, E)) = \Theta_\Pi \cdot H_c^2(\Pi, E).$$

Having fixed ζ_∞ , the period Θ_Π is well defined modulo $E^\times$ and rescaling ζ_∞ rescales Θ_Π .

4.8.2 Integral structures on cohomology

Let $v \mid p$ be a place of E , with completion E_v . The $\mathcal{O}_{E_v}$ -points of $V_{\lambda, \mathbb{Z}}^\vee$ carry an action of $\mathrm{GL}_3(\mathbb{Z}_p)$, giving a local system $\mathcal{V}_\lambda^\vee(\mathcal{O}_{E_v})$ of $\mathcal{O}_{E_v}$ -modules on $Y^{\mathrm{GL}_3}(\mathcal{U})$ as in Section 4.4. The group $H_c^2(Y^{\mathrm{GL}_3}(\mathcal{U}), \mathcal{V}_\lambda^\vee(\mathcal{O}_{E_v}))$ is a finitely generated $\mathcal{O}_{E_v}$ -module, and its quotient by torsion is a lattice in $H_c^2(Y^{\mathrm{GL}_3}(\mathcal{U}), \mathcal{V}_\lambda^\vee(E_v))$. As in [LW26, §3.3.3], given a finite set $\{W_i\} \subset \mathcal{W}(\Pi_f, E)^\mathcal{U}$ and a finite extension $L/\mathbb{Q}_p$ into which E embeds, the classes

$$\phi_\Pi(W_i)/\Theta_\Pi \in H_c^2(Y^{\mathrm{GL}_3}(\mathcal{U}), \mathcal{V}_\lambda^\vee(L))$$

can, after rescaling Θ_Π by a p -adic unit times an element of $E^\times$, be normalised to lie in

$$H_c^2(Y^{\mathrm{GL}_3}(\mathcal{U}), \mathcal{V}_\lambda^\vee(\mathcal{O}_L))/\{\text{torsion}\}.$$

Such a normalisation depends on $\{W_i\}$, which will be harmless as our test data is fixed once and for all in Section 9.

5 Eisenstein series and Eisenstein classes for GL_2

In this section we recall, largely following [LW26, §5], the theory of Eisenstein series attached to adelic Schwartz functions, the motivic Eisenstein classes of Beilinson, and their p -adic interpolation due to Kings. Throughout this section, symmetric spaces are for GL_2 unless indicated otherwise.

5.1 Schwartz functions

For a field K of characteristic 0, let $\mathcal{S}(\mathbb{A}_f^2, K)$ denote the space of locally constant, compactly supported functions $\mathbb{A}_f^2 \rightarrow K$, and $\mathcal{S}_0(\mathbb{A}_f^2, K)$ the subspace of Φ with $\Phi(0, 0) = 0$. The group $\mathrm{GL}_2(\mathbb{A}_f)$ acts on these spaces by $(g \cdot \Phi)(v) = \Phi(vg)$.

If χ is a Dirichlet character, let R_χ denote the projection to the $\hat{\chi}^{-1}$ -isotypical component:

$$R_\chi(\Phi_f)(x, y) := \int_{a \in \hat{\mathbb{Z}}^\times} \hat{\chi}(a) \Phi_f(ax, ay) d^\times a.$$

5.2 Eisenstein series

Let $\Phi = \Phi_\infty \Phi_f \in \mathcal{S}(\mathbb{A}^2, \mathbb{C})$ (Schwartz at infinity) and let $\hat{\chi}$ be a finite-order Hecke character. For $g \in \mathrm{GL}_2(\mathbb{A})$ and $\Re(s) \gg 0$, define the Godement–Siegel section and Eisenstein series

$$f^\Phi(g; \hat{\chi}, s) := \|\det g\|^s \int_{\mathbb{A}^\times} \Phi((0, a)g) \hat{\chi}(a) \|a\|^{2s} d^\times a, \quad E^\Phi(g; \hat{\chi}, s) := \sum_{\gamma \in B_{\mathrm{GL}_2}(\mathbb{Q}) \backslash \mathrm{GL}_2(\mathbb{Q})} f^\Phi(\gamma g; \hat{\chi}, s),$$

which have meromorphic continuation to all s . The series is analytic if $\Phi(0, 0) = \hat{\Phi}(0, 0) = 0$ or if $\hat{\chi}$ is ramified at some finite place. Classically, for $\Phi_f \in \mathcal{S}_{(0)}(\mathbb{A}_f^2, K)$ and τ in the upper half-plane one sets

$$E_{\Phi_f}^{j+2}(\tau; s) := \frac{\Gamma(s + \frac{j+2}{2})}{(-2\pi i)^{j+2} \pi^{s - \frac{j+2}{2}}} \sum_{(m, n) \in \mathbb{Q}^2 \setminus (0, 0)} \frac{\Phi_f(m, n) y^{s - \frac{j+2}{2}}}{(m\tau + n)^{j+2} |m\tau + n|^{2s - j - 2}},$$

extended to a function $E_{\Phi_f}^{j+2}(g; s)$ on $\mathrm{GL}_2(\mathbb{A})$ by $\mathrm{GL}_2(\mathbb{A}_f)$ -equivariance in Φ_f and the standard automorphy recipe at infinity [Wei71, §I], and we further set $E_{\Phi_f}^{j+2}(g; \chi, s) := E_{R_\chi(\Phi_f)}^{j+2}(g; s)$. Taking

$$\Phi_\infty^{j+2}(x, y) := 2^{-1-j} (x + iy)^{j+2} \exp(-\pi(x^2 + y^2)), \quad (13)$$

the adelic and classical series are related by $E^\Phi(g; \hat{\chi}, s) = \|\det g\|^s E_{\Phi_f}^{j+2}(g; \chi, s)$ for $\Phi = \Phi_\infty^{j+2} \Phi_f$ [LW26, Proposition 5.2, Cor. 5.3]. We will be interested exclusively in the special value $s = -j/2$, and set

$$\mathcal{E}_{\Phi_f}^{j+2, \chi}(g) := E_{\Phi_f}^{j+2}(g; \chi, -\frac{j}{2}) = \|\det g\|^{j/2} E^{R_\chi(\Phi)}(g; \hat{\chi}, -\frac{j}{2}), \quad (14)$$

which depends $\mathrm{GL}_2(\mathbb{A}_f)$ -equivariantly on Φ_f and transforms as a vector in the principal series $I(\|\cdot\|^{-1/2}, \hat{\chi}^{-1} \cdot \|\cdot\|^{j+1/2})$.

5.3 Betti–Eisenstein classes

For $j \geq 0$ let $\mathcal{V}_{(0,-j)}$ denote the local system on Y^{GL_2} attached to the GL_2 -representation $V_{(0,-j)}^{\mathrm{GL}_2}$ of highest weight $(0, -j)$, as in Section 4.4. The main geometric input to our construction is the following theorem of Beilinson [Bei86], in the form given in [LW26, Theorem 5.5, Cor. 5.6] (see also [LSZ21, §7]): for every $j \geq 0$ and every level U , there is a canonical $\mathrm{GL}_2(\mathbb{A}_f)$ -equivariant map, the *Betti–Eisenstein symbol*,

$$\mathcal{S}_{(0)}(\mathbb{A}_f^2, \mathbb{Q})^U \rightarrow H^1(Y^{\mathrm{GL}_2}(U), \mathcal{V}_{(0,-j)}(\mathbb{Q})), \quad \Phi_f \mapsto \mathrm{Eis}_{\Phi_f}^j,$$

compatible with Hecke operators and change of level, whose image under the Betti–de Rham comparison is the differential form $-\mathcal{E}_{\Phi_f}^{j+2}(g)(dz)^{\otimes j} d\tau$. Here the notation $\mathcal{S}_{(0)}$ means $\mathcal{S}_0$ if $j = 0$ and $\mathcal{S}$ otherwise.

These classes are not integral, but they admit integral ‘ c -smoothings’: for $c > 1$ coprime to 6 and to the level data, there are classes ${}_c\mathrm{Eis}_{\Phi_f}^j \in H^1(Y^{\mathrm{GL}_2}(U), \mathcal{V}_{(0,-j)}(\mathbb{Z}))$, compatible with Hecke operators and change of level, satisfying

$${}_c\mathrm{Eis}_{\Phi_f}^j = \left(c^2 - c^{-j} \binom{c}{c}\right) \mathrm{Eis}_{\Phi_f}^j \quad (15)$$

after inverting c , where $\binom{c}{c} \in \mathrm{GL}_2(\mathbb{A}_f^{(c)})$, see [LW26, Theorem 5.7].

5.4 The Eisenstein–Iwasawa class

Fix a prime-to- p level $\mathcal{U}^{\mathrm{GL}_2, (p)}$ and a prime-to- p Schwartz function $\Phi^{(p)} \in \mathcal{S}(\mathbb{A}_f^{(p), 2}, \mathbb{Z})^{\mathcal{U}^{\mathrm{GL}_2, (p)}}$, and let c be coprime to $6p$ and to all level data. For $t \geq 1$ let $\Phi_{p,t} := \mathrm{ch}((0, 1) + p^t \mathbb{Z}_p^2)$ and $\Phi_{f,t} := \Phi^{(p)} \Phi_{p,t}$, which is stable under $\mathcal{U}_t^{\mathrm{GL}_2}$ and compatible with the trace maps as t varies. The resulting classes with $j = 0$ assemble into the *Eisenstein–Iwasawa class*

$${}_c\mathcal{EI}_{\Phi^{(p)}} := \left({}_c\mathrm{Eis}_{\Phi_{f,t}}^0\right)_{t \geq 1} \in H_{\mathrm{Iw}}^1(Y^{\mathrm{GL}_2}(\mathcal{U}_{\infty}^{\mathrm{GL}_2}), \mathbb{Z}_p) := \varprojlim_t H^1(Y^{\mathrm{GL}_2}(\mathcal{U}_t^{\mathrm{GL}_2}), \mathbb{Z}_p).$$

This Iwasawa cohomology group admits moment maps

$$\mathrm{mom}_t^j : H_{\mathrm{Iw}}^1(Y^{\mathrm{GL}_2}(\mathcal{U}_{\infty}^{\mathrm{GL}_2}), \mathbb{Z}_p) \rightarrow H^1(Y^{\mathrm{GL}_2}(\mathcal{U}_t^{\mathrm{GL}_2}), \mathcal{V}_{(0,-j)}(\mathbb{Z}_p))$$

for each $j \geq 0$ and $t \geq 1$ (see [LSZ21, §9.1]), and the fundamental theorem of Kings on the Λ -adic interpolation of Eisenstein classes (see [Kin13], or [KLZ17] for a summary in the present notation) states that

$$\mathrm{mom}_t^j({}_c\mathcal{EI}_{\Phi^{(p)}}) = {}_c\mathrm{Eis}_{\Phi_{f,t}}^j \quad \text{for all } j \geq 0, t \geq 1. \quad (16)$$

All of the above is compatible with the action of $\mathrm{GL}_2(\mathbb{A}_f^{(pc)})$.

If $\Phi^{(p)}$ lies in the $\hat{\chi}$ -eigenspace for the action of the centre for some Dirichlet character χ of prime-to- p conductor, then it can be shown (cf. [LW26, §5.3.3]) that the class

$$\mathcal{EI}_{\Phi^{(p)}} := (c^2 - c^{-\mathbf{j}} \chi^{-1}(c))^{-1} \otimes {}_c\mathcal{EI}_{\Phi^{(p)}} \in H_{\mathrm{Iw}}^1(Y^{\mathrm{GL}_2}(\mathcal{U}_{\infty}^{\mathrm{GL}_2}), \mathbb{Z}_p) \otimes_{\Lambda} \mathrm{Frac}(\Lambda)$$

is independent of the choice of c , where $\mathbf{j} : \mathbb{Z}_p^{\times} \rightarrow \Lambda$ is the universal character. Moreover, if χ is non-trivial then $\mathcal{EI}_{\Phi^{(p)}} \in H_{\mathrm{Iw}}^1(Y^{\mathrm{GL}_2}(\mathcal{U}_{\infty}^{\mathrm{GL}_2}), \mathbb{Z}_p) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ and if $\chi \not\equiv 1 \pmod{p}$ then this class is actually integral.

6 p -adic distributions

In this section we recall the spaces of p -adic distributions and the interpolation results needed to construct our p -adic L -function.

6.1 Spaces of analytic functions and distributions

Let $L/\mathbb{Q}_p$ be a complete extension.

Definition 6.1.1. We define the following spaces:

- For an integer $a \geq 0$ let $LP^{[0,a]}(\mathbb{Z}_p^\times, L)$ denote the space of locally polynomial functions on $\mathbb{Z}_p^\times$ of degree $\leq a$.
- Let $\mathcal{D}_{\text{alg}}^{[0,a]}(\mathbb{Z}_p^\times, L)$ denote the L -dual of $LP^{[0,a]}(\mathbb{Z}_p^\times, L)$.
- Let $LA(\mathbb{Z}_p^\times, L)$ denote the space of locally analytic functions $f : \mathbb{Z}_p^\times \rightarrow L$. We equip $LA(\mathbb{Z}_p^\times, L)$ with the inductive limit topology given by the Banach subspaces $LA_h(\mathbb{Z}_p^\times, L)$ of functions which are analytic on discs of radius p^{-h} , for $h \geq 1$, equipped with the following norm: For $f \in LA_h(\mathbb{Z}_p^\times, L)$ given on the disc $z + p^h\mathbb{Z}_p$, $z \in \mathbb{Z}_p^\times$, by $f(z + p^h\mathbb{Z}_p) = \sum_{k \geq 0} a_{z,k} z^k$, define

$$\|f\|_{LA_h} := \sup_{z \in \mathbb{Z}_p^\times} \sup_k |a_{z,k}|.$$

- Let $\mathcal{D}_{\ell a}(\mathbb{Z}_p^\times, L)$ denote the continuous L -dual of $LA(\mathbb{Z}_p^\times, L)$, referred to as the space of *locally analytic distributions* $\mu : LA(\mathbb{Z}_p^\times, L) \rightarrow L$ on $\mathbb{Z}_p^\times$. It has the structure of a Fréchet space defined by the family of norms $\|f\|_{LA_h}$. We equip $\mathcal{D}_{\ell a}(\mathbb{Z}_p^\times, L)$ with the action of $\delta \in \mathbb{Z}_p^\times$ given by

$$\int_{\mathbb{Z}_p^\times} f(x) (\delta \cdot \mu)(x) = \int_{\mathbb{Z}_p^\times} f(\delta^{-1}x) \mu(x),$$

where for $f \in LA(\mathbb{Z}_p^\times, L)$ we use the standard notation $\int_{\mathbb{Z}_p^\times} f(x) \mu(x) := \mu(f)$.

For $h \in \mathbb{R}_{\geq 0}$ there is a filtration $\mathcal{D}_h(\mathbb{Z}_p^\times, L) \subset \mathcal{D}_{\ell a}(\mathbb{Z}_p^\times, L)$ by Banach spaces, defined in [Col10, II.3.1]. For our purposes we are interested in the following properties of these spaces.

Proposition 6.1.2. Let $h \in \mathbb{R}_{\geq 0}$ and let $\mu \in \mathcal{D}_h(\mathbb{Z}_p^\times, L)$. Then

1. Suppose $\mu_{p^\infty} \subset L$ and that for all Dirichlet characters χ of p -power conductor and for all $0 \leq j \leq [h]$ we have

$$\int_{\mathbb{Z}_p^\times} \chi(x) x^j \mu(x) = 0,$$

then $\mu \equiv 0$.

2. There is $C \in \mathbb{R}$ such that for all $z \in \mathbb{Z}_p^\times$, $n \geq 1, j \geq 0$

$$\left| \int_{z+p^n\mathbb{Z}_p} \left(\frac{x-z}{p^n} \right)^j \mu(x) \right| \leq Cp^{hn}.$$

3. Suppose for some $a \geq [h]$, all $0 \leq j \leq a$, and $n \geq 0$ we have $\mu_{\text{alg}} \in \mathcal{D}_{\text{alg}}^{[0,a]}(\mathbb{Z}_p^\times, L)$ satisfying

$$\left| \int_{z+p^n\mathbb{Z}_p} \left(\frac{x-z}{p^n} \right)^j \mu_{\text{alg}}(x) \right| \leq Cp^{hn}.$$

Then there is a unique $\mu \in \mathcal{D}_h(\mathbb{Z}_p^\times, L)$ whose restriction to $LP^{[0,a]}(\mathbb{Z}_p^\times, L)$ is μ_{alg} .

Proof. Parts (2) and (3) for distributions on $\mathbb{Z}_p$ are [Col10, Théorème II.3.2]; we can deduce the result for $\mathbb{Z}_p^\times$ by considering μ as a distribution on $\mathbb{Z}_p$ supported on $\mathbb{Z}_p^\times$. For the first part [Col10, Proposition I.3.4] implies that Dirichlet characters χ of p -power conductor are a basis for the space of locally constant functions on $\mathbb{Z}_p^\times$ (as for parts (2) and (3), the Proposition of *op. cit.* is stated for locally polynomial functions on $\mathbb{Z}_p$, but we can deduce the result for $\mathbb{Z}_p^\times$ by restriction) and thus that functions of the form $x \mapsto \chi(x)x^j$ for $0 \leq j \leq [h]$ are a basis for $LP^{[0,[h]]}(\mathbb{Z}_p^\times, L)$. Then parts (2) and (3) say that μ is uniquely determined by its restriction to $LP^{[0,[h]]}(\mathbb{Z}_p^\times, L)$, which is identically zero by assumption. $\square$

7 Constructing the distribution

We now assemble the ingredients of the previous sections into a p -adic distribution attached to a small slope P_1 -refined RACAR. Throughout this section we fix:

- A RACAR Π of $\mathrm{GL}_3(\mathbb{A})$ of weight $\lambda = (a, 0, -a)$, together with a regular unramified P_1 -refinement σ_p , with $\alpha_p = \sigma_p(p)$ and $r = r(\tilde{\Pi})$ as in Proposition 3.3.9. We write

$$\alpha := p^{a+1}\alpha_p, \quad h := v_p(\alpha) \in [0, 2a + 2]$$

for the normalised $U_{p,1}$ -eigenvalue and its slope. We further assume that $\tilde{\Pi}$ satisfies the small slope hypothesis $h < a + 1$, noting that this enters only in Theorem 7.5.1.

- A prime-to- p Schwartz function $\Phi^{(p)} \in \mathcal{S}(\mathbb{A}_f^{(p),2}, \mathbb{Z})$ and prime-to- p levels $\mathcal{U}^{G,(p)} = \mathcal{U}^{\mathrm{GL}_3,(p)} \times \prod_{\ell \neq p} \mathbb{Z}_\ell^\times$, $\mathcal{U}^{H,(p)} = \mathcal{U}^{G,(p)} \cap H(\mathbb{A}_f^{(p)})$ and $\mathcal{U}^{\mathrm{GL}_2,(p)} = \mathrm{pr}_{\mathrm{GL}_2}(\mathcal{U}^{H,(p)})$, where $\mathrm{pr}_{\mathrm{GL}_2}$ is the projection to the GL_2 -factor of H , such that $\Phi^{(p)}$ is fixed by $\mathcal{U}^{H,(p)}$. We assume the resulting levels are neat, shrinking $\mathcal{U}^{\mathrm{GL}_3,(p)}$ if necessary.
- An auxiliary even Dirichlet character η_2 of conductor D prime to p , valued in $\mathcal{O}_L$, with $\mathcal{U}^{G,(p)}(1, D) \subset \ker(\hat{\eta}_2 \circ \nu_2)$. The role of this character is solely in removing the smoothing factor (Theorem 7.5.1(2)), and its only effect on the final result is on the period $\Omega_{\tilde{\Pi}}^-$.
- An integer $c > 1$ coprime to $6pD$ and to all the level data.

7.1 The classes $z_n^{[a,j]}$

Recall the Eisenstein–Iwasawa class ${}_c\mathcal{E}\mathcal{I}_{\Phi(p)} \in H_{\mathrm{Iw}}^1(Y^{\mathrm{GL}_2}(\mathcal{U}_\infty^{\mathrm{GL}_2}), \mathbb{Z}_p)$ of Section 5.4. Pulling back along the natural maps $\tilde{Y}^H \xrightarrow{f} Y^H \rightarrow Y^{\mathrm{GL}_2}$ of (7) yields a class. We define two maps

$$\iota_{a,j,n} : H_{\mathrm{Iw}}^1(Y^{\mathrm{GL}_2}(\mathcal{U}_\infty^{\mathrm{GL}_2}(1, D)), \mathbb{Z}_p) \longrightarrow H^3(Y^{\mathrm{GL}_3}(\mathcal{V}_n^{\mathrm{GL}_3}), (\gamma_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \otimes \mathrm{Maps}(\Delta_{1,n} \times \Delta_2, \mathcal{O}) \quad (17)$$

and

$$\iota_{a,j,n,t}^L : H^1(Y^{\mathrm{GL}_2}(\mathcal{U}_t^{\mathrm{GL}_2}), \mathcal{V}_{(0,-j;0)}(\mathbb{Z}_p)) \longrightarrow H^3(Y^{\mathrm{GL}_3}(\mathcal{V}_n^{\mathrm{GL}_3}), \gamma_\lambda(\mathbb{Z}_p)) \otimes \mathrm{Maps}(\Delta_{1,n} \times \Delta_2, \mathcal{O}), \quad (18)$$

the first given by the composition

$$\begin{aligned} H_{\mathrm{Iw}}^1(Y^{\mathrm{GL}_2}(\mathcal{U}_\infty^{\mathrm{GL}_2}), \mathbb{Z}_p) &\rightarrow H_{\mathrm{Iw}}^1(\tilde{Y}^H(Q_{H,\infty,n}^0(p^n, D)), \mathbb{Z}_p) \\ &\xrightarrow{\cup f^{a,j}} H_{\mathrm{Iw}}^1(\tilde{Y}^H(Q_{H,\infty,n}^0(p^n, D)), \iota^*(\gamma_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \\ &\xrightarrow{\iota_*} H_{\mathrm{Iw}}^3(Y^G(Q_{H,\infty,n}^0(p^n, D)), (\gamma_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \\ &\xrightarrow{[u\tau^n]_{*,j}} H_{\mathrm{Iw}}^3(Y^G((u\tau^n)^{-1}Q_{H,\infty,n}^0(p^n, D)(u\tau^n)), (\gamma_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \\ &\xrightarrow{\mathrm{pr}_n} H^3(Y^G(\mathcal{V}_n^G(p^n, D)), (\gamma_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \\ &\xrightarrow{\Psi_j} H_{\mathrm{Iw}}^3(Y^G(\mathcal{V}_n^G), (\gamma_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \otimes \mathrm{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p), \end{aligned}$$

where we use the convention that when an H -subgroup is the level at p in G -Iwasawa cohomology, the tame level is taken to be $\mathcal{U}^{G,(p)}$. The second map given for any $t \geq \max\{n, r\}$ by

$$\begin{aligned}
H^1(Y^{\text{GL}_2}(\mathcal{U}_t^{\text{GL}_2}), \mathcal{V}_{(0,-j)}(\mathbb{Z}_p)) &\rightarrow H^1\left(\tilde{Y}^H(Q_{H,t,n}^0(p^n, D)), \mathcal{V}_{(0,-j;0)}(\mathbb{Z}_p)\right) \\
&\xrightarrow{T_j} H^1\left(\tilde{Y}^H(Q_{H,t,n}^0(p^n, D)), \mathcal{V}_{(j,0;-j)}(\mathbb{Z}_p)\right) \\
&\xrightarrow{\text{br}^{[a,j]}} H^1(\tilde{Y}^H(Q_{H,t,n}^0(p^n, D)), \iota^* \mathcal{V}_\lambda(\mathbb{Z}_p)) \\
&\xrightarrow{\iota_*} H^3(Y^G(u\mathcal{U}_n^G(p^n, D)u^{-1}), \mathcal{V}_\lambda(\mathbb{Z}_p)) \\
&\xrightarrow{[u\tau^n]_{*,0}} H^3(Y^G(\mathcal{V}_n^G(p^n, D)), \mathcal{V}_\lambda(\mathbb{Z}_p)) \\
&\xrightarrow{\Psi_0} H^3(Y^G(\mathcal{V}_n^G), \mathcal{V}_\lambda(\mathbb{Z}_p)) \otimes \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p)
\end{aligned}$$

The map $\iota_{a,j,n}$ is useful for proving p -adic interpolation, whereas the map $\iota_{a,j,n,t}^L$ is useful for relations to L -values. We will routinely abuse notation and also use $\iota_{a,j,n,t}^L$ to refer to the composition following initial pullback from Y^{GL_2} .

Lemma 7.1.1. *The following diagram commutes.*

$$\begin{array}{ccc}
H_{\text{Iw}}^1\left(\tilde{Y}^H(\mathcal{U}_\infty^H(1, D)), \mathbb{Z}_p\right) & \xrightarrow{\iota_{a,j,n}} & H_{\text{Iw}}^3\left(Y^G(\mathcal{V}_n^G), \left(\mathcal{V}_\lambda \otimes \nu_1^{-j}\right)(\mathbb{Z}_p)\right) \otimes \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p) \\
\downarrow \text{mom}_{j,t}^H & & \downarrow T_j \otimes 1 \\
H^1\left(\tilde{Y}^H(\mathcal{U}_t^H(1, D)), \mathcal{V}_{(0,-j;0)}(\mathbb{Z}_p)\right) & \xrightarrow{\iota_{a,j,n,t}^L} & H^3(Y^G(\mathcal{V}_n^G), \mathcal{V}_\lambda(\mathbb{Z}_p)) \otimes \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p)
\end{array}$$

In particular, this composition is independent of t .

Proof. We first recall that we have a diagram

$$\begin{array}{ccc}
\mathbb{Z}_p & \xrightarrow{a} & V_{\lambda, \mathbb{Z}_p} \otimes \nu_1^{-j} \\
\downarrow b & & \parallel \\
V_{(0,-j;0), \mathbb{Z}_p}^H & \xrightarrow{c} & V_{\lambda, \mathbb{Z}_p} \otimes \nu_1^{-j}
\end{array} \tag{19}$$

where the a and b are $Q_H^0(\mathbb{Z}_p)$ -equivariant, and c is $H(\mathbb{Z}_p)$ -equivariant. Let $v_j \in V_{(0,-j;0), \mathbb{Z}_p}^H$ be a choice of highest weight vector, then we normalise a to send $1 \mapsto f^{a,j}$, b to send $1 \mapsto v_j$, and c to send $v_j \mapsto f^{a,j}$.

Let $z \in H_{\text{Iw}}^1\left(\tilde{Y}^H(Q_{H,\infty,n}^0(p^n, D)), \mathbb{Z}_p\right)$ and write

$$\text{pr}_{t,n} : H_{\text{Iw}}^1\left(\tilde{Y}^H(Q_{H,\infty,n}^0(p^n, D)), \mathcal{V}_{(0,-j;0)}^H(\mathbb{Z}_p)\right) \rightarrow H_{\text{Iw}}^1\left(\tilde{Y}^H(Q_{H,t,n}^0(p^n, D)), \mathcal{V}_{(0,-j;0)}^H(\mathbb{Z}_p)\right)$$

be the projection map, so that $\text{mom}_{j,t,n}^H = \text{pr}_{t,n} \circ (- \cup v_j)$. Since $T_j \otimes 1$ is an isomorphism with inverse $T_{-j} \otimes 1$, it suffices to show that $\iota_{a,j,n} = (T_{-j} \otimes 1) \circ \iota_{a,j,n,t}^L \circ \text{mom}_{j,t}^H$. Write $\mathcal{P}_{n,j} := [u\tau^n]_{*,j} \circ \iota_*$. Writing $w = \text{mom}_{j,t,n}^H(z)$, we then have

$$\begin{aligned}
(T_{-j} \otimes 1) \circ \iota_{a,j,n,t}^L(w) &= (T_{-j} \otimes 1) \circ \left(\Psi_0 \circ \mathcal{P}_{n,0} \circ \text{br}^{[a,j]}(w \cup \mathbb{S}_j)\right) \\
&= \Psi_j \circ T_{-j} \circ \mathcal{P}_{n,0} \circ \text{br}^{[a,j]}(w \cup \mathbb{S}_j) \\
&= \Psi_j \circ \mathcal{P}_{n,j} \left(\text{br}^{[a,j]}(w \cup \mathbb{S}_j) \cup \mathbb{S}_{-j}\right) \\
&= \Psi_j \circ \mathcal{P}_{n,j} \left(\left(\text{br}^{[a,j]} \otimes \nu_1^{-j}\right)(w)\right),
\end{aligned}$$

where $\text{br}^{[a,j]} \otimes \nu_1^{-j}$ is the morphism of sheaves induced by the map of H -representations $V_{(0,-j;0)}^H \rightarrow V_\lambda \otimes \nu_1^{-j}$, the second equality uses that cupping with $\mathbb{S}_j$ is equivariant for the normalised action of τ^n , and the

third equality uses Lemma 4.7.3, the projection formula and the fact that $\iota^* \mathbb{S}_{-j} = \mathbb{S}_{-j}$. In other words, $(T_{-j} \otimes 1) \circ \iota_{a,j,n,t}^L$ is given by the composition:

$$\begin{aligned} H^1(Y^{\text{GL}_2}(\mathcal{U}_t^{\text{GL}_2}), \mathcal{V}_{(0,-j)}(\mathbb{Z}_p)) &\rightarrow H^1(\tilde{Y}^H(Q_{H,t,n}^0(p^n, D)), \mathcal{V}_{(0,-j;0)}(\mathbb{Z}_p)) \\ &\xrightarrow{\text{br}^{[a,j]} \otimes \nu_1^{-j}} H^1(\tilde{Y}^H(Q_{H,t,n}^0(p^n, D)), \iota^*(\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \\ &\xrightarrow{\iota_*} H_{\text{Iw}}^3(Y^G(u\mathcal{U}_n^G(p^n, D)u^{-1}), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \\ &\xrightarrow{[u\tau^n]_*} H^3(Y^G(\mathcal{V}_n^G(p^n, D)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \\ &\xrightarrow{\Psi_j} H^3(Y^G(\mathcal{V}_n^G), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \otimes \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p) \end{aligned}$$

Since both compositions defining $\iota_{a,j,n}$ and $(T_{-j} \otimes 1) \circ \iota_{a,j,n,t}^L$ terminate in the same map, it suffices to show that

$$[u\tau^n]_{*,j} \circ \iota_* \circ \text{br}^{[a,j]} \otimes \nu_1^{-j} \circ \text{pr}_{t,n}(z \cup v_j) = \text{pr}_n \circ [u\tau^n]_{*,j} \circ \iota_*(z \cup \iota^* f^{a,j}), \quad (20)$$

where we recall that $\varphi_{(0,-j;0)}^{hw}$ is the section of $\mathcal{V}_{(0,-j;0)}^H$ corresponding to $v_j \in V_{(0,-j;0)}^H$. The expression $z \cup \iota^* f^{a,j}$ is the inverse limit over t of the mod p^t expressions

$$\text{pr}_{t,n}(z) \cup (\iota^* \varphi_t^{a,j}) \in H^1(\tilde{Y}^H(Q_{H,t,n}^0(p^n, D)), \iota^*(\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)/p^t).$$

On the other hand, the composition $\text{br}^{[a,j]} \otimes \nu_1^{-j} \circ \text{pr}_{t,n}(z \cup v_j)$ reduces modulo p^t to the expression $\text{br}^{[a,j]} \otimes \nu_1^{-j} \circ (\text{pr}_{t,n}(z) \cup (\varphi_{v_j} \bmod p^t))$, so we deduce from (19) that

$$\text{br}^{[a,j]} \otimes \nu_1^{-j} \circ (\text{pr}_{t,n}(z \cup v_j)) \equiv \text{pr}_{t,n}(z \cup \iota^* f^{a,j}) \bmod p^t.$$

The map $\iota_* : H^1(\tilde{Y}^H(Q_{H,t,n}^0(p^n, D)), \iota^*(\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \rightarrow H_{\text{Iw}}^3(Y^G(u\mathcal{U}_n^G(p^n, D)u^{-1}), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p))$ factors as

$$\begin{aligned} &H^1(\tilde{Y}^H(Q_{H,t,n}^0(p^n, D)), \iota^*(\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \\ &\rightarrow H_{\text{Iw}}^3(Y^G(Q_{H,t,n}^0(p^n, D)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \\ &\rightarrow H^3(Y^G(u\mathcal{U}_n^G(p^n, D)u^{-1}), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)). \end{aligned}$$

and it's easy to see that there's a commutative diagram

$$\begin{array}{ccc} H_{\text{Iw}}^3(Y^G(Q_{H,\infty,n}^0(p^n, D)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) & \xrightarrow{[u\tau^n]_*} & H_{\text{Iw}}^3(Y^G((u\tau^n)^{-1}Q_{H,\infty,n}^0(p^n, D)(u\tau^n)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \\ \downarrow & & \downarrow \\ H^3(Y^G(u\mathcal{U}_n^G(p^n, D)u^{-1}), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) & \xrightarrow{[u\tau^n]_*} & H^3(Y^G(\mathcal{V}_n^G(p^n, D)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \end{array}$$

from which we deduce that

$$[u\tau^n]_* \circ \iota_* \circ \text{br}^{[a,j]} \otimes \nu_1^{-j} \circ \text{pr}_{t,n}(z \cup v_j) \equiv \text{pr}_n \circ [u\tau^n]_* \circ (\iota_*(z \cup \iota^* f^{a,j})) \bmod p^t,$$

i.e. as elements of $H^3(Y^G(\mathcal{V}_n^G(p^n, D)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)/p^t)$. Taking the limit over t , we obtain the result. $\square$

Definition 7.1.2. For $0 \leq j \leq a$ and $n \geq 1$, set

$$c_z^{[a,j]}(\Phi^{(p)}) := \iota_{a,j,n}(c_{\mathcal{E}\mathcal{I}_{\Phi^{(p)}}}) \in H^3(Y^{\text{GL}_3}(\mathcal{V}_n^{\text{GL}_3}), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \otimes \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p).$$

Further define ‘unsmoothed’ versions

$$z_n^{[a,j]}(\Phi^{(p)}) := (T_{-j} \otimes 1) \iota_{a,j,n,t}^L \left(\text{Eis}_{\Phi_{f,t}}^j \right)$$

using the unsmoothed Eisenstein classes $\text{Eis}_{\Phi_{f,t}}^j$ recalled in Section 5.3.

Note that by Lemma 7.1.1 we have ${}_cz_n^{[a,j]}(\Phi^{(p)}) \cup \mathbb{S}_j = \iota_{a,j,n,t}^L \left({}_c\text{Eis}_{\Phi_{f,t}}^j \right)$

7.2 The norm relation

We show that the classes ${}_cz_n^{[a,j]}(\Phi^{(p)}) \cup \mathbb{S}_j$ satisfy a norm relation as we vary n . Let

$$\text{Norm}_{\Delta_{1,n}}^{\Delta_{1,n+1}} : \text{Maps}(\Delta_{1,n+1} \times \Delta_2, \mathbb{Z}_p) \rightarrow \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p)$$

be the map satisfying $\text{Norm}_{\Delta_{1,n}}^{\Delta_{1,n+1}}(f)((\delta_{1,n}, \delta_2)) = \sum_{(\varepsilon_{1,n}, \varepsilon_2)} f((\varepsilon_{1,n+1}, \varepsilon_2))$, where the sum is over all $(\varepsilon_{1,n}, \varepsilon_2) \in \Delta_{1,n+1} \times \Delta_2$ that map to $(\delta_{1,n}, \delta_2)$ in Δ_{n+1} .

Proposition 7.2.1. *For all $n \geq 1$ we have*

$$\left[([1]_{\mathcal{V}_n}^{n+1})_* \otimes \text{Norm}_{\Delta_{1,n}}^{\Delta_{1,n+1}} \right] \left({}_cz_{n+1}^{[a,j]}(\Phi^{(p)}) \cup \mathbb{S}_j \right) = (U'_p \otimes 1) \cdot {}_cz_n^{[a,j]}(\Phi^{(p)}) \cup \mathbb{S}_j$$

in $H^3(Y^{\text{GL}_3}(\mathcal{V}_n^{\text{GL}_3}), \mathcal{V}_\lambda(\mathbb{Z}_p)) \otimes \text{Maps}(\Delta_{1,n} \times \Delta_2, \mathbb{Z}_p)$, and similarly for $z_n^{[a,j]}$.

Proof. This proof follows the general shape of [Loe21, Proposition 4.5.2, §4.6, §5.2.3], and [LW26, Theorem 6.10]. We sketch the argument. Recalling that

$${}_cz_n^{[a,j]}(\Phi^{(p)}) \cup \mathbb{S}_j = \iota_{a,j,n,t}^L \left({}_c\text{Eis}_{\Phi_{f,t}}^j \right)$$

and writing $\tilde{z}_n \in H^3(Y^G(\mathcal{U}_n^G(p^n, D)), \mathcal{V}_\lambda(\mathbb{Z}_p))$ for the classes before the τ^n -translation in the composition defining $\iota_{a,j,n,t}^L$, one has

$$\left([1]_{\mathcal{U}_{n+1}^G(p^n, D)}^{\mathcal{U}_{n+1}^G(p^{n+1}, D)} \right)_* (\tilde{z}_{n+1}) = \left([1]_{\mathcal{U}_n^G(p^n, D)}^{\mathcal{U}_{n+1}^G(p^n, D)} \right)^* (\tilde{z}_n). \quad (21)$$

Indeed, for $t \geq \max\{n+1, r\}$ we have a diagram

$$\begin{array}{ccccc} \tilde{Y}^H(Q_{H,t,n+1}^0(p^{n+1}, D)) & \xrightarrow{[u] \circ \iota} & \tilde{Y}^H(\mathcal{U}_{n+1}^G(p^{n+1}, D)) & & \\ \parallel & & \downarrow & & \\ \tilde{Y}^H(Q_{H,t,n+1}^0(p^{n+1}, D)) & \xrightarrow{[u] \circ \iota} & Y^G(\mathcal{U}_{n+1}^G(p^n, D)) & \xrightarrow{[\tau]} & Y^G(\tau^{-1}\mathcal{U}_{n+1}^G(p^n, D)\tau) \\ \downarrow & & \downarrow & & \downarrow \\ \tilde{Y}^H(Q_{H,t,n}^0(p^n, D)) & \xrightarrow{[u] \circ \iota} & Y^G(\mathcal{U}_n^G(p^n, D)) & & Y^G(\mathcal{U}_n^G(p^n, D)). \end{array} \quad (22)$$

The bottom left square is cartesian: to show this it suffices to show that the map of cosets

$$Q_{H,t,n}^0(p^n, D)/Q_{H,t,n+1}(p^{n+1}, D) \rightarrow \mathcal{U}_n^G(p^n, D)/\mathcal{U}_{n+1}^G(p^n, D)$$

induced by $u \circ \iota$ is a bijection. Since both vertical maps have degree p^2 by Lemma 4.3.1(1) it suffices to show injectivity which is equivalent to showing that $Q_{H,t,n}(p^n, D) \cap u\mathcal{U}_{n+1}^G(p^n, D)u^{-1} = Q_{H,t,n+1}(p^{n+1}, D)$ which is easily seen. The rightmost horseshoe defines the unnormalised Hecke operator $p^{-a}U'_p$. It follows that

$$[\tau]_{\mathcal{U}_n^G(p^n, D),*}^{\mathcal{U}_{n+1}^G(p^{n+1}, D)}(\tilde{z}_{n+1}) = p^{-a}U'_p \cdot \tilde{z}_n,$$

where $[\tau]_{\mathcal{U}_n^G(p^n, D)}^{\mathcal{U}_{n+1}^G(p^{n+1}, D)} : Y^G(\mathcal{U}_{n+1}^G(p^{n+1}, D)) \rightarrow Y^G(\mathcal{U}_n^G(p^n, D))$ is the map induced by pushforward along the top and right arrows in the horseshoe. Applying $[\tau^n]_*$ we obtain

$$[1]_{\mathcal{V}_n^G(p^n, D),*}^{\mathcal{V}_{n+1}^G(p^{n+1}, D)}([\tau^{n+1}]_{*,0}\tilde{z}_{n+1}) = U'_p \cdot [\tau^n]_{*,0}\tilde{z}_n.$$

These maps are all compatible with Ψ_0 and so we get the result. $\square$

7.3 Pairing with a finite-slope eigenclass

Let $\varphi_f = \otimes \varphi_\ell \in \Pi_f$ be such that:

- (a) φ_p is the P_1 -refined newvector: φ_p is fixed by $\mathcal{U}_{1,p}^{(P_1)}(p^r)$ and $U_{p,1}\varphi_p = \alpha \varphi_p$;
- (b) $W_{\varphi_f} \in \mathcal{W}(\Pi_f, E)$ is algebraic, fixed by $\mathcal{U}^{\text{GL}_3, (p)}$ away from p .

Enlarging L if necessary, and normalising Θ_Π as in Section 4.8, we obtain a class

$$\phi = \phi_{\varphi_f} := \phi_\Pi(W_{\varphi_f})/\Theta_\Pi \in H_c^2(Y^{\text{GL}_3}(\mathcal{U}), \mathcal{V}_\lambda^\vee(\mathcal{O}_L)) / \{\text{torsion}\},$$

and we write ϕ_n for its pullback to level $\mathcal{V}_n^{\text{GL}_3} \subset \mathcal{U}$.

Definition 7.3.1. For $n \geq 1$ and $0 \leq j \leq a$, let $C = |\text{vol}(\mathcal{U}^{G, (p)}(1, D) \cap H(\mathbb{A}_f^{(p)}))|_p$ and define

$${}_c\tilde{\Xi}_n^{[a; j]}(\phi, \Phi^{(p)}) := \text{vol}(\mathcal{U}^{G, (p)}(1, D) \cap H(\mathbb{A}_f^{(p)})) \cdot (1 \otimes e_{\eta_2}) \left\langle {}_c z_n^{[a; j]}(\Phi^{(p)}) \cup \mathbb{S}_j, \phi_n \right\rangle_{\mathcal{V}_n} \in C^{-1} \cdot \text{Maps}(\Delta_{1, n}, \mathcal{O}),$$

where $\langle -, - \rangle_{\mathcal{V}_n}$ is the Poincaré pairing (10) applied at level $\mathcal{V}_n$, $e_{\eta_2} : \text{Maps}(\Delta_2, \mathcal{O}_L) \rightarrow \mathcal{O}_L$ is the map $f \mapsto \sum_{x \in \Delta_2} \eta_2(x) f(x)$, and the volume is with respect to dh_f . We further set

$${}_c\Xi_n^{[a; j]} := \alpha^{-n} \cdot {}_c\tilde{\Xi}_n^{[a; j]} \in C^{-1} p^{-hn} \mathcal{O}_L[\Delta_{1, n}],$$

where $h := v_p(\alpha)$, and define $\tilde{\Xi}_n^{[a; j]}, \Xi_n^{[a; j]}$ (with L -coefficients) analogously using $z_n^{[a; j]}$.

Remark 7.3.2. The map $(1 \otimes e_{\eta_2}) \circ \iota_{a, j, n, t}^L$ recovers the construction of [LW26, §6], with the same tame levels and branching-map normalisations. Here we identify

$$\text{Maps}(\Delta_{1, n}, \mathcal{O}_L) \cong \mathcal{O}_L[\Delta_{1, n}], \quad f \mapsto \sum_{x \in \Delta_{1, n}} f(x)[x].$$

In particular,

$$(1 \otimes e_{\eta_2}) \iota_{a, j, n, t}^L \left({}_c\text{Eis}_{\Phi_f, t}^j \right) = {}_c\xi_{n, \text{LW}}^{[a; j]},$$

where the right-hand side denotes the class constructed in *op. cit.* Indeed, Lemma 4.3.1 identifies the source level with that used in their construction. Their coefficient twist, given in the rational local-system model by multiplication by $\|\nu_1(h)\|^{-j}$, becomes multiplication by

$$\|\nu_1(h)\|^{-j} \nu_1(h_p)^{-j} = \mathbb{S}_j(h)$$

under the comparison isomorphism (8). It therefore agrees with T_j . Moreover, on the union of connected components labelled by $(x, y) \in \Delta_{1, n} \times \Delta_2$, applying Ψ_0 and then $1 \otimes e_{\eta_2}$ records the contribution with weight $\eta_2(y)[x]$. This is precisely the effect of cupping with $[\nu_{1, (n)}]$ and $\hat{\eta}_2 \circ \nu_2$ in *op. cit.* The translation by $u\tau^n$ preserves these component labels. The remaining operations are the same branching map, pushforward and translation, with total normalisation p^{an} , incorporated here in $[u\tau^n]_{*, 0}$. This proves the claimed identification.

The same comparison holds for the unsmoothed classes. Consequently, pairing with the same automorphic cohomology class and applying the same volume and eigenvalue normalisations gives the finite-level quantities computed in [LW26, §7].

As in [LW26, Proposition 6.11], the volume normalisation makes ${}_c\tilde{\Xi}_n^{[a; j]}$ independent of the choice of $\mathcal{U}^{G, (p)}$ fixing the data.

Corollary 7.3.3. For all $n \geq 1$ we have $\text{Norm}_{\Delta_{1, n}}^{\Delta_{1, n+1}} \left({}_c\Xi_{n+1}^{[a; j]} \right) = {}_c\Xi_n^{[a; j]}$.

Proof. Write $z_n := {}_c z_n^{[a; j]}(\Phi^{(p)}) \cup \mathbb{S}_j$. Since $\phi_{n+1} = ([1]_{\mathcal{V}_n^{n+1}})^* \phi_n$, adjointness of pullback and pushforward under (10) gives

$$\text{Norm} \langle {}_c z_{n+1}, \phi_{n+1} \rangle = \langle ([1])_* \otimes \text{Norm} \rangle {}_c z_{n+1}, \phi_n \rangle = \langle (U'_p \otimes 1) {}_c z_n, \phi_n \rangle = \langle {}_c z_n, U_p \phi_n \rangle = \alpha \langle {}_c z_n, \phi_n \rangle,$$

using Proposition 7.2.1, the adjointness of U_p and U'_p , and (a). Multiplying by $\alpha^{-(n+1)}$ gives the claim. $\square$

7.4 Congruences between twists

We now prove that the constructed classes $c\Xi_n^{[a,j]}$ satisfy congruences as j varies. Recall that we identify

$$\Delta_{1,n} := \mathbb{Q}_{>0}^\times \backslash \mathbb{A}_f^\times / \nu_1(\mathcal{V}_n(p^n)) \cong (\mathbb{Z}/p^n\mathbb{Z})^\times$$

using the conventions of [LW26] that a uniformiser ϖ_ℓ at $\ell \neq p$ is sent to $\ell \bmod p^n$. In particular, $z \in \mathbb{Z}_p^\times \subset \mathbb{A}_f^\times$ is sent to $z^{-1} \bmod p^n$. For all $n \geq 1, 0 \leq j \leq a$ recall that in Lemma 4.7.3 we defined a section $\mathbb{S}_{j,n} \in H^0(Y_G(\mathcal{V}_n(p^n, D)), \nu_1^j/p^n)$ whose image under the isomorphism to $\text{Maps}((\mathbb{Z}/p^n\mathbb{Z})^\times \times \Delta_2, \mathcal{O})$ is $(z, \delta) \mapsto z^j$ and is the reduction of $\mathbb{S}_j$ modulo p^n . For the proof of the following Proposition we abuse notation somewhat and write res_n^{nk} for the pullback

$$H_{\text{Iw}}^3(Y^G(K(p^n, D)), V) \rightarrow H_{\text{Iw}}^3(Y^G(K(p^{nk}, D)), V)$$

for any closed subgroup K and any appropriate local system V , the specifics being dictated by context, as well as for the natural pullback map

$$\text{Maps}\left((\mathbb{Z}/p^n\mathbb{Z})^\times, \mathbb{Z}_p\right) \rightarrow \text{Maps}\left((\mathbb{Z}/p^{nk}\mathbb{Z})^\times, \mathbb{Z}_p\right),$$

and any linear extension thereof.

Proposition 7.4.1. *For all $0 \leq k \leq a$ and $n \geq 1$ the classes $\text{res}_n^{nk} \left(c\Xi_n^{[a,j]}(\Phi^{(p)}) \right) \in \text{Maps}\left((\mathbb{Z}/p^{nk}\mathbb{Z})^\times, L\right)$ satisfy*

$$\left\| p^{-kn} \sum_{j=0}^k (-1)^j \binom{k}{j} \text{tw}_{-j} \left(\text{res}_n^{nk} \left(c\Xi_n^{[a,j]}(\Phi^{(p)}) \right) \right) \right\| \leq Cp^{hn},$$

where $\|\cdot\|$ is the sup norm, and when $k = 0$ we replace res_n^{nk} with the identity map.

Proof. We first note that for $k = 0$ the statement of the proposition follows immediately from the integrality of the classes $c\Xi_n^{[a,j]}$. Set $\mathcal{R}_{n,j} = \text{pr}_n \circ [u\tau^n]_{*,j} \circ \iota_*$. Note first that Lemma 4.7.3 (2) gives us

$$\begin{aligned} \text{tw}_{-j} \circ \text{res}_n^{nk} \left\langle c\Xi_n^{[a,j]}(\Phi^{(p)}) \cup \mathbb{S}_j, \phi_n \right\rangle_{\mathcal{V}_n} &= \left\langle \text{res}_n^{nk} \left(c\Xi_n^{[a,j]}(\Phi^{(p)}) \right) \cup \mathbb{S}_j \otimes \text{tw}_{-j}, \phi_n \right\rangle_{\mathcal{V}_n} \\ &\equiv \langle \Psi_0 (\text{res}_n^{nk} \circ \mathcal{R}_{n,j} (c\mathcal{E}\mathcal{I}_{\Phi^{(p)}} \cup f^{a,j}) \cup \varphi_{j,nk}), \phi_n \rangle_{\mathcal{V}_n} \bmod p^{nk} \end{aligned}$$

We now show that the classes

$$z_{n,j} = [u\tau^n]_{*,j} \circ \iota_* (c\mathcal{E}\mathcal{I}_{\Phi^{(p)}} \cup \iota^* f^{a,j}) \in H^3 \left(Y^G((u\tau^n)^{-1} Q_H^0(p^n, D)(u\tau^n)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p) \right)$$

satisfy

$$\sum_{j=0}^k (-1)^j \binom{k}{j} \text{res}_n^{nk} (z_{n,j}) \cup \varphi_{j,nk} \equiv 0 \bmod p^{nk} \quad (23)$$

for all $1 \leq k \leq a$. Indeed, recalling that we defined $[u\tau^n]_{*,j} = p^{(a-j)n} [u\tau^n]_*$ as a map

$$H_{\text{Iw}}^3(Y^G(Q_H^0(p^n, D)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p)) \rightarrow H^3 \left(Y^G((\tau^n u)^{-1} Q_H^0(p^n, D)(\tau^n u)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p) \right),$$

we have (all modulo p^{nk}), letting $z := c\mathcal{E}\mathcal{I}_{\Phi^{(p)}}$,

$$\begin{aligned} \text{res}_n^{nk} (z_{n,j}) \cup \varphi_{j,nk} &= \text{res}_n^{nk} ([u\tau^n]_{*,j} \circ \iota_* (z \cup \iota^* f^{a,j})) \cup \varphi_{j,nk} \\ &= \text{res}_n^{nk} ([u\tau^n]_{*,j} \circ (\iota_* (z) \cup f^{a,j})) \cup \varphi_{j,nk} \\ &= p^{(a-j)n} \text{res}_n^{nk} ([u\tau^n]_* \circ (\iota_* (z) \cup f^{a,j})) \cup \varphi_{j,nk} \\ &= \text{res}_n^{nk} \left([u\tau^n]_* (\iota_* (z)) \cup p^{(a-j)n} [u\tau^n]_* f^{a,j} \right) \cup \varphi_{j,nk} \\ &= \text{res}_n^{nk} ([u\tau^n]_* (\iota_* (z))) \cup p^{(a-j)n} [u\tau^n]_* f^{a,j} \cup \varphi_{j,nk} \\ &= \text{res}_n^{nk} ([u\tau^n]_* (\iota_* (z))) \cup p^{an} [u\tau^n]_* (f^{a,j} \cup \varphi_{j,nk}) \\ &= \text{res}_n^{nk} ([u\tau^n]_* (\iota_* (z))) \cup [u\tau^n]_{*,0} (f^{a,j} \cup \varphi_{j,nk}) \end{aligned}$$

the section $[u\tau^n]_{*,0} (f^{a,j} \cup \varphi_{j,nk})$ is induced by the integral vector $\tau^{-n} * (u^{-1} \cdot \nu_1^j f^{a,j}) \in V_{\lambda, \mathbb{Z}_p}$ and thus by Proposition 2.4.4 the classes $z_{n,j}$ satisfy the congruence (23). We claim that the following diagram commutes

$$\begin{array}{ccc} H^3_{\text{Iw}} \left(Y^G((u\tau^n)^{-1} Q_{H,\infty,n}^0(p^n, D)(u\tau^n)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p) \right) & \xrightarrow{\text{pr}_n} & H^3 \left(Y^G(\mathcal{V}_n(p^n, D)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p) \right) \\ \downarrow \text{res}_n^{nk} & & \downarrow \text{res}_n^{nk} \\ H^3 \left(Y^G((u\tau^n)^{-1} Q_{H,\infty,n}^0(p^{nk}, D)(u\tau^n)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p) \right) & \xrightarrow{\text{pr}_{nk}} & H^3 \left(Y^G(\mathcal{V}_n(p^{nk}, D)), (\mathcal{V}_\lambda \otimes \nu_1^{-j})(\mathbb{Z}_p) \right) \end{array} \quad (24)$$

Indeed, if $K_t \subset G(\mathbb{Q}_p)$ is a family of open compact subgroups satisfying $\cap_t K_t = ((u\tau^n)^{-1} Q_{H,\infty,n}^0(u\tau^n))_p$ and $K_t(p^n, D) \subset \mathcal{V}_n(p^n, D)$ (adjoining the tame level $\mathcal{U}^{G,(p)}$ to K_t and leaving it implicit) then the diagram

$$\begin{array}{ccc} Y^G(K_t(p^{nk}, D)) & \xrightarrow{[1]_{\mathcal{V}_n(p^{nk}, D)}^{K_t(p^{nk}, D)}} & Y^G(\mathcal{V}_n(p^{nk}, D)) \\ \downarrow & & \downarrow \\ Y^G(K_t(p^n, D)) & \xrightarrow{[1]_{\mathcal{V}_n(p^n, D)}^{K_t(p^n, D)}} & Y^G(\mathcal{V}_n(p^n, D)) \end{array}$$

is cartesian as ν_1 identifies

$$K_t(p^n, D)/K_t(p^{nk}, D) \cong (1 + p^n \mathbb{Z}_p) / (1 + p^{nk} \mathbb{Z}_p) \cong \mathcal{V}_n(p^n, D)/\mathcal{V}_n(p^{nk}, D).$$

This gives a commutative diagram

$$\begin{array}{ccc} H^3(Y^G(K_t(p^n, D)), \mathcal{V}_\lambda(\mathbb{Z}_p)) & \xrightarrow{\text{pr}_n} & H^3(Y^G(\mathcal{V}_n(p^n, D)), \mathcal{V}_\lambda(\mathbb{Z}_p)) \\ \downarrow \text{res}_n^{nk} & & \downarrow \text{res}_n^{nk} \\ H^3(Y^G(K_t(p^{nk}, D)), \mathcal{V}_\lambda(\mathbb{Z}_p)) & \xrightarrow{\text{pr}_n^{(nk)}} & H^3(Y^G(\mathcal{V}_n(p^{nk}, D)), \mathcal{V}_\lambda(\mathbb{Z}_p)) \end{array}$$

and we deduce commutativity of (24) by taking the limit over t . Moreover, its easy to see that we have a commutative diagram

$$\begin{array}{ccc} H^3(Y^G(\mathcal{V}_n(p^n, D)), \mathcal{V}_\lambda(\mathbb{Z}_p)) & \xrightarrow{\Psi_0} & H^3(Y^{\text{GL}_3}(\mathcal{V}_n^G), \mathcal{V}_\lambda(\mathbb{Z}_p)) \otimes \text{Maps}\left((\mathbb{Z}/p^n \mathbb{Z})^\times \times \Delta_2, \mathbb{Z}_p\right) \\ \downarrow \text{res}_n^{nk} & & \downarrow 1 \otimes \text{res}_n^{nk} \\ H^3(Y^G(\mathcal{V}_n(p^{nk}, D)), \mathcal{V}_\lambda(\mathbb{Z}_p)) & \xrightarrow{\Psi_0} & H^3(Y^{\text{GL}_3}(\mathcal{V}_n^G), \mathcal{V}_\lambda(\mathbb{Z}_p)) \otimes \text{Maps}\left((\mathbb{Z}/p^{nk} \mathbb{Z})^\times \times \Delta_2, \mathbb{Z}_p\right) \end{array}$$

From the above discussion and Lemma 4.7.3 we deduce that

$$\Psi_0 \circ \text{pr}_n \circ (\text{res}_n^{nk}(z_{n,j}) \cup \varphi_{j,nk}) = (1 \otimes \text{tw}_{-j}) \text{res}_n^{nk}(\iota_{a,j,n}(z)) \cup \mathbb{S}_j,$$

where as before $\cup \mathbb{S}_j$ is acting as the identity on $\text{Maps}\left((\mathbb{Z}/p^{nk} \mathbb{Z})^\times, \mathbb{Z}/p^{nk} \mathbb{Z}\right)$. Setting $v := \text{vol}\left(\mathcal{U}^{G,(p)}(1, D) \cap H(\mathbb{A}_f^{(p)})\right)$ we conclude that

$$\text{tw}_{-j} \circ \text{res}_n^{nk} \left(v^{-1} {}_c \tilde{\Xi}_n^{[a,j]} \right) = (1 \otimes e_{\eta_2}) \langle \Psi_0 \circ \text{pr}_n \circ (\text{res}_n^{nk}(z_{n,j}) \cup \varphi_{j,nk}), \phi_n \rangle \text{ mod } p^{nk}$$

from which it follows immediately from (23) that

$$C \cdot \sum_{j=0}^k (-1)^j \binom{k}{j} \text{tw}_{-j} \left(\text{res}_n^{nk} \left({}_c \tilde{\Xi}_n^{[a;j]} \left(\phi, \Phi^{(p)} \right) \right) \right) \equiv 0 \text{ mod } p^{nk}$$

as elements of $\text{Maps}\left((\mathbb{Z}/p^{nk}\mathbb{Z})^\times, \mathcal{O}_L/p^{nk}\right)$, which we can equivalently write as

$$\left\| \sum_{j=0}^k (-1)^j \binom{k}{j} \text{tw}_{-j} \left(\text{res}_n^{nk} \left({}_c\tilde{\Xi}_n^{[a;j]} \left(\phi, \Phi^{(p)} \right) \right) \right) \right\| \leq Cp^{-nk}.$$

It is immediate that the normalised classes satisfy

$$\left\| \alpha^n \sum_{j=0}^k (-1)^j \binom{k}{j} \text{tw}_{-j} \left(\text{res}_n^{nk} \left({}_c\Xi_n^{[a;j]} \left(\phi, \Phi^{(p)} \right) \right) \right) \right\| \leq Cp^{-nk},$$

which rearranges to

$$\left\| p^{nk} \sum_{j=0}^k (-1)^j \binom{k}{j} \text{tw}_{-j} \left(\text{res}_n^{nk} \left({}_c\Xi_n^{[a;j]} \left(\phi, \Phi^{(p)} \right) \right) \right) \right\| \leq Cp^{hn}.$$

□

7.5 Tempered distributions

Set

$$\Theta_c := c^2 - (\eta_2 \omega_\Pi)(c)^{-1} [c]^{-1} \in \mathcal{O}_L[[\mathbb{Z}_p^\times]],$$

As in [LW26, (6.3), Proposition 6.13], the smoothed and non-smoothed classes are related by

$${}_c\Xi_n^{[a;j]} = (c^2 - c^{-j}(\eta_2 \omega_\Pi)(c)^{-1} [c]^{-1}) \cdot \Xi_n^{[a;j]}. \quad (25)$$

Theorem 7.5.1. *Suppose $\tilde{\Pi}$ has small slope, $h < a + 1$. Then:*

1. *There is a unique ${}_c\mu_{\tilde{\Pi}} = {}_c\mu_{\tilde{\Pi}}(\phi, \Phi^{(p)}) \in \mathcal{D}_h(\mathbb{Z}_p^\times, L)$ such that for all $n \geq 1$, all Dirichlet characters χ of conductor dividing p^n , and all $0 \leq j \leq a$,*

$$\int_{\mathbb{Z}_p^\times} \chi(x) x^j {}_c\mu_{\tilde{\Pi}}(x) = \sum_{\gamma \in (\mathbb{Z}/p^n\mathbb{Z})^\times} \chi(\gamma) {}_c\Xi_n^{[a;j]}(\gamma).$$

2. *Suppose η_2 is chosen so that $\eta_2 \omega_\Pi$ is not congruent modulo $\mathfrak{m}_L$ to any character of p -power conductor. Then for suitable c the element Θ_c is a unit in $\mathcal{O}_L[[\mathbb{Z}_p^\times]]$, and*

$$\mu_{\tilde{\Pi}}(\phi, \Phi^{(p)}) := \Theta_c^{-1} * {}_c\mu_{\tilde{\Pi}}(\phi, \Phi^{(p)}) \in \mathcal{D}_h(\mathbb{Z}_p^\times, L),$$

where $*$ denotes convolution of distributions, is independent of c and satisfies

$$\int \chi(x) x^j \mu_{\tilde{\Pi}} = \sum_{\gamma \in (\mathbb{Z}/p^n\mathbb{Z})^\times} \chi(\gamma) \Xi_n^{[a;j]}(\gamma)$$

for all χ, j, n as in (1).

Proof. If we let $\mathcal{C}(\mathbb{Z}_p^\times, \mathbb{Z}_p)$ denote the space of continuous functions taking values in $\mathbb{Z}_p$ and $\mathcal{C}(\mathbb{Z}_p, L) := \mathcal{C}(\mathbb{Z}_p^\times, \mathbb{Z}_p) \hat{\otimes} L$ the space of continuous functions taking values in L , then we have compatible injective restriction maps

$$\text{res} : \text{Maps}((\mathbb{Z}/p^n\mathbb{Z})^\times, \mathbb{Z}_p) \rightarrow \mathcal{C}(\mathbb{Z}_p^\times, \mathbb{Z}_p)$$

for all $n \geq 1$ and twisting maps $\text{tw}_j : \mathcal{C}(\mathbb{Z}_p^\times, \mathbb{Z}_p) \rightarrow \mathcal{C}(\mathbb{Z}_p^\times, \mathbb{Z}_p)$ sending $z \mapsto f(z)$ to $z \mapsto z^j f(z)$ which are compatible modulo p^n with the previously defined maps $\text{tw}_j : \text{Maps}((\mathbb{Z}/p^n\mathbb{Z})^\times, \mathbb{Z}/p^n\mathbb{Z}) \rightarrow \text{Maps}((\mathbb{Z}/p^n\mathbb{Z})^\times, \mathbb{Z}/p^n\mathbb{Z})$. We equip $\mathcal{C}(\mathbb{Z}_p^\times, \mathbb{Z}_p)$ with the sup norm $\|f\| = \sup_{z \in \mathbb{Z}_p^\times} |f|$ and $\mathcal{C}(\mathbb{Z}_p^\times, L)$ with its

natural linear extension. We will abuse notation and write ${}_c\Xi_n^{[a;j]}$ for its restriction to $\mathcal{C}(\mathbb{Z}_p^\times, C^{-1}\mathcal{O}_L)$ and similarly ${}_c\Xi_n^{[a;j]}$ for the analogous element of $\mathcal{C}(\mathbb{Z}_p^\times, L)$. There is a unique element ${}_c\mu_{\tilde{\Pi}, \text{alg}} \in \mathcal{D}_{\text{alg}}^{[0,a]}(\mathbb{Z}_p^\times, L)$ satisfying

$$\int_{z+p^n\mathbb{Z}_p} x^j {}_c\mu_{\tilde{\Pi}, \text{alg}}(x) = {}_c\Xi_n^{[a;j]}(z) \in L$$

for all $0 \leq j \leq a$, which is a well-defined algebraic distribution by the norm relation of Corollary 7.3.3. We have for $z \in \mathbb{Z}_p^\times$, $n \geq 1$

$$\begin{aligned} \int_{z+p^n\mathbb{Z}_p} \left(\frac{x-z}{p^n}\right)^k {}_c\mu_{\tilde{\Pi}, \text{alg}}(x) &= p^{-nk} \sum_{j=0}^k \binom{k}{j} (-1)^{k-j} z^{k-j} \int_{z+p^n\mathbb{Z}_p} x^j {}_c\mu_{\tilde{\Pi}, \text{alg}}(x) \\ &= z^k p^{-nk} \sum_{j=0}^k \binom{k}{j} (-1)^{k-j} \text{tw}_{-j} \left({}_c\Xi_n^{[a;j]} \right) (z) \end{aligned}$$

By Proposition 7.4.1 we thus have

$$\left| \int_{z+p^n\mathbb{Z}_p} \left(\frac{x-z}{p^n}\right)^k {}_c\mu_{\tilde{\Pi}, \text{alg}}(x) \right| \leq Cp^{hn}$$

for all $0 \leq k \leq a$ and $n \geq 1$, thus by Proposition 6.1.2 (using the small slope assumption $h < a+1$) there is ${}_c\mu_{\tilde{\Pi}} \in \mathcal{D}_h(\mathbb{Z}_p^\times, L)$ satisfying ${}_c\mu_{\tilde{\Pi}}|_{LP^{[0,a]}(\mathbb{Z}_p^\times, L)} = {}_c\mu_{\tilde{\Pi}, \text{alg}}$. Let χ be a Dirichlet character of conductor p^n . Then we have

$$\begin{aligned} \int_{\mathbb{Z}_p^\times} \chi(x) x^j {}_c\mu_{\tilde{\Pi}}(x) &= \sum_{z \in (\mathbb{Z}/p^n\mathbb{Z})^\times} \chi(z) \int_{z+p^n\mathbb{Z}_p} x^j {}_c\mu_{\tilde{\Pi}}(x) \\ &= \sum_{z \in (\mathbb{Z}/p^n\mathbb{Z})^\times} \chi(z) {}_c\Xi_n^{[a;j]}(z), \end{aligned}$$

as required. Part (2) follows immediately. $\square$

8 Values of the distribution as Rankin–Selberg integrals

In this section we compute the values $\int \eta(x) x^j {}_c\mu_{\tilde{\Pi}}(x)$ as global Rankin–Selberg integrals and evaluate the resulting local zeta integrals. By Theorem 7.5.1 these values equal $\eta(\Xi_n^{[a;j]})$, and by Remark 7.3.2 the classes $\Xi_n^{[a;j]}$ are those of [LW26, §6]; the computation of this section is therefore essentially identical to [LW26, §§7–9], and we give a summary adapted to our notation, referring to *op. cit.* for the details. We stress that these computations are insensitive to the slope: the only point in *op. cit.* where ordinarity is used is the boundedness of the measure, which we have replaced by Theorem 7.5.1, and in Lemma 8.8 of *op. cit.* which generalises immediately to the small slope setting, given here as Lemma 8.2.3.

8.1 Archimedean data and the global integral

Recall the generator ζ_∞ of (6). Choosing bases $\{\delta'_1, \dots, \delta'_5\}$ of $(\mathfrak{gl}_3/\mathfrak{k}_{3,\infty}^\circ)^\vee$ and $\{v_\alpha\}$ of $V_\lambda^\vee(\mathbb{C})$, there are vectors $\varphi_{\infty, r, s, \alpha} \in \Pi_\infty$ with

$$\zeta_\infty = \sum_{r,s=1}^5 \sum_{\alpha} (\delta'_r \wedge \delta'_s) \otimes \varphi_{\infty, r, s, \alpha} \otimes v_\alpha,$$

and we set $\varphi_{r, s, \alpha} := \varphi_{\infty, r, s, \alpha} \otimes \varphi_f \in \Pi$. Similarly, the Eisenstein differential is described by a basis $\{\delta_1, \delta_2\}$ of $(\mathfrak{gl}_2/\mathfrak{k}_{2,\infty}^\circ)^\vee$ and a basis $\{w_\beta^{[j]}\}$ of $V_{(0,-j)}^{\text{GL}_2}(\mathbb{C})$; we normalise $\iota^*(\delta'_i) = \delta_i$ for $i = 1, 2, 3$ (extending to a basis $\{\delta_1, \delta_2, \delta_3\}$ on $\tilde{\mathcal{H}}_H$) and $\iota^*(\delta'_i) = 0$ for $i = 4, 5$, and fix the Haar measure dh_∞ corresponding to $\delta_1 \wedge \delta_2 \wedge \delta_3$, normalised so that $\text{vol}(K_{H,\infty}^\circ) = 1$. Finally, the branching maps of Section 2.4 induce pairings

$$\langle -, - \rangle_{a,j} : V_\lambda^\vee \times V_{(0,-j)}^{\text{GL}_2} \rightarrow \mathbb{G}_a, \quad (\mu, v) \mapsto \mu(\text{br}^{a,j}\text{-image of } v),$$

as in [LW26, (4.4)], integral for our choices of lattices.

Definition 8.1.1. Let χ_1, χ_2 be Dirichlet characters and $s_1, s_2 \in \mathbb{C}$. For $\varphi \in \Pi$ and $\Phi \in \mathcal{S}(\mathbb{A}^2, \mathbb{C})$, set

$$\mathcal{Z}(\varphi, \Phi; \chi_1, \chi_2, s_1, s_2) := \int_{\mathrm{GL}_2(\mathbb{Q}) \backslash \mathrm{GL}_2(\mathbb{A})} \varphi(\iota(g, 1)) E^\Phi(g; \hat{\chi}_2, s_2) \hat{\chi}_1(\det g) \|\det g\|^{s_1 - \frac{1}{2}} dg.$$

For $\Re(s_1) \gg 0$ this factors as an Euler product $\mathcal{Z} = \prod_v Z_v(W_{\varphi, v}, \Phi_v; \hat{\chi}_{1, v}, \hat{\chi}_{2, v}, s_1, s_2)$ of local Jacquet–Piatetski-Shapiro–Shalika zeta integrals [JPSS83], cf. [LW26, (7.8), Definition 8.1].

Proposition 8.1.2. Let η be a Dirichlet character of p -power conductor, and let $n = \max(1, v_p(\mathrm{cond} \eta))$. Then for $0 \leq j \leq a$,

$$\int_{\mathbb{Z}_p^\times} \eta(x) x^j \mu_{\bar{\Pi}}(\phi_{\varphi_f}, \Phi^{(p)})(x) = \frac{\mathrm{vol}(\mathcal{U}_{n, p}^H)^{-1}}{p^n \alpha_p^n \Theta_\Pi} \sum_{r, s, t} \varepsilon_{rst} \sum_{\alpha, \beta} \left\langle v_\alpha, w_\beta^{[j]} \right\rangle_{a, j} \mathcal{Z}(u\tau^n \cdot \varphi_{r, s, \alpha}, \Phi; \eta, \omega_\Pi \eta \eta_2, \frac{1-j}{2}, -\frac{j}{2}),$$

where $\varepsilon_{231} = 1 = -\varepsilon_{132}$ and $\varepsilon_{rst} = 0$ otherwise, $\Phi = \Phi_\infty^{j+2} \Phi^{(p)} \Phi_{p, R}$, and $\mathrm{vol}(\mathcal{U}_{n, p}^H)^{-1} = p^{2R+2n}(1-p^{-1})(1-p^{-2})$ with $R = \max(n, r(\bar{\Pi}))$.

Proof. By Theorem 7.5.1 the left-hand side is $e_\eta(\Xi_n^{[a; j]})$. Via Remark 7.3.2, this is precisely the quantity computed in [LW26, §7.1–7.3]: one expresses the cup products as integrals of differential forms over $\tilde{Y}^H$, sums over the connected components against η and η_2 , unfolds the Eisenstein series, and performs the change of variables collapsing the central integration into the projection $R_{\omega_\Pi \eta \eta_2}$ on the Schwartz data; the volume computation is Lemma 4.3.1(1). We refer to *op. cit.* for the details, which apply verbatim. $\square$

8.2 Local zeta integrals

The local integrals at finite places $v \neq p$ where all data is unramified take the value 1 after normalising by the local L -factors, and at the remaining places away from p they may be made identically 1 by a suitable finite linear combination of test data:

Theorem 8.2.1 (Jacquet–Piatetski-Shapiro–Shalika). Let v be a finite place, π a generic irreducible representation of $\mathrm{GL}_3(\mathbb{Q}_v)$, and χ_1, χ_2 smooth characters of $\mathbb{Q}_v^\times$. The normalised integral

$$\tilde{Z}(W, \Phi; \chi_1, \chi_2, s_1, s_2) := \frac{Z(W, \Phi; \chi_1, \chi_2, s_1, s_2)}{L(\pi \times \chi_1, s_1 + s_2 - \frac{1}{2}) L(\pi \times \chi_1 \chi_2^{-1}, s_1 - s_2 + \frac{1}{2})}$$

is a polynomial in $\ell^{\pm s_1 \pm s_2}$. The ideal generated by these polynomials for varying (W, Φ) is the unit ideal, and there are finite collections $\{W_i\}, \{\Phi_i\}$ defined over $\bar{\mathbb{Q}}$ with $\sum_i \tilde{Z}(W_i, \Phi_i; \chi_1, \chi_2, s_1, s_2) = 1$ identically. If all data is unramified and normalised spherical, then $\tilde{Z} = 1$.

Proof. See [JPSS83] and [LW26, Theorem 8.2]. $\square$

At $v = p$ we use the following evaluation, which is [LW26, Theorem 8.7]. We emphasise that the proof in *op. cit.* is a purely local computation with the refined newvector W^α of Proposition 3.3.9 and makes no use of ordinarity; it is valid for any regular unramified P_1 -refinement of finite slope.

Theorem 8.2.2. Let η be of conductor p^{n_1} , $n = \max(1, n_1)$, $R = \max(n, r)$, and take the test data $W = u\tau^n \cdot W^\alpha$ and $\Phi_R = \mathrm{ch}((0, 1) + p^R \mathbb{Z}_p^2)$. Then

$$Z(W, \Phi_R; \hat{\eta}_p, \hat{\omega}_{\Pi, p} \hat{\eta}_p \hat{\eta}_{2, p}, \frac{1-j}{2}, -\frac{j}{2}) = \frac{\alpha_p^n}{p^{2R+n}(1-p^{-1})(1-p^{-2})} \cdot e_p(\Pi_p \times \eta_p, -j) \cdot \frac{\mathcal{E}_0 \cdot L(\Pi_p \times \omega_{\Pi, p}^{-1} \eta_{2, p}^{-1}, 1)}{\varepsilon(\Pi_p \times \omega_{\Pi, p}^{-1} \eta_{2, p}^{-1}, 1)},$$

where, with $\mathcal{E}_s := L(\sigma'_p \times \sigma_p \eta_{2, p}, s) / L(\Pi_p^\vee \times \omega_{\Pi, p} \eta_{2, p}, s)$, the factor $\mathcal{E}_0$ is as in the following lemma.

Lemma 8.2.3. *The factor $\mathcal{E}_0$ equals 1 if the induction of $\sigma_p \times \sigma'_p$ to $\mathrm{GL}_3(\mathbb{Q}_p)$ is reducible, and equals $1 - \omega_\Pi(p)\eta_2(p)\alpha_p^{-1}$ otherwise. If $\tilde{\Pi}$ has small slope then $\mathcal{E}_0$ is a p -adic unit.*

Proof. The identification of $\mathcal{E}_0$ is [LW26, Lemma 8.8, Rem. 8.9], using $L(\Pi_p^\vee \times \omega_{\Pi,p}, s) = L(\wedge^2 \Pi_p, s)$. In the irreducible, small slope gives $v_p(\alpha_p^{-1}) > 0$ (Definition 3.3.4), while $\omega_\Pi(p)\eta_2(p)$ is a root of unity, so $\equiv 1 \pmod{\mathfrak{m}_L}$. $\square$

At $v = \infty$, by [LW26, Lemma 8.10] (following [JPSS83] and [Mah00, §1]) the local integral is a polynomial multiple of the expected pair of archimedean L -factors, and combining the sums over (r, s, t, α, β) into a single archimedean quantity we set

$$\tilde{e}_\infty(\zeta_\infty \times \eta_\infty, -j) := \sum_{r,s,t} \varepsilon_{rst} \sum_{\alpha,\beta} \left\langle v_\alpha, w_\beta^{[j]} \right\rangle_{a,j} Z_\infty(W_{\infty,r,s,\alpha}, \Phi_\infty^{j+2}; \hat{\eta}_\infty, \hat{\omega}_{\Pi,\infty} \hat{\eta}_\infty, \frac{1-j}{2}, -\frac{j}{2}),$$

which is non-zero by the non-vanishing results of Kasten–Schmidt [KS13] and Sun [Sun17], and depends only on Π_∞ , ζ_∞ and j .

8.3 Interpolation, and the factor at infinity

Combining Proposition 8.1.2 with Theorems 8.2.1 and 8.2.2 (for test data trivialising the away-from- p integrals, as fixed in Notation 9.0.1 below), the volume factors and powers of α_p cancel and we obtain: for all $(-j, \eta) \in \mathrm{Crit}_p^-(\Pi)$,

$$\int_{\mathbb{Z}_p^\times} \eta(x) x^j \mu_{\tilde{\Pi}}(x) = \tilde{e}_\infty(\zeta_\infty \times \eta_\infty, -j) e_p(\Pi_p \times \eta_p, -j) \cdot \frac{L^{(p)}(\Pi \times \eta, -j)}{\Omega_{\tilde{\Pi}}}, \quad (26)$$

where

$$\Omega_{\tilde{\Pi}}^- := \frac{\Theta_\Pi \cdot \varepsilon(\Pi_p \times (\omega_{\Pi,p} \eta_{2,p})^{-1}, 1)}{\mathcal{E}_0 \cdot L(\Pi \times (\omega_\Pi \eta_2)^{-1}, 1)} \in \mathbb{C}^\times \quad (27)$$

(well defined and non-zero: the L -value is non-critical-central and non-vanishing, and $\mathcal{E}_0 \neq 0$ by Lemma 8.2.3).

It remains to identify $\tilde{e}_\infty$ with the Coates–Perrin–Riou factor e_∞ of Section 3.2.5. The quantity $\tilde{e}_\infty(\zeta_\infty \times \eta_\infty, -j)$ is precisely the same archimedean factor as in [LW26], depending only on $(\Pi_\infty, \zeta_\infty, j)$, and in particular is insensitive to p and to the slope. By [LW26, §9.3–9.5] where it is shown, by comparison with symmetric square p -adic L -functions at the prime 3 for auxiliary self-dual representations sharing the given Π_∞ , that the $\tilde{e}_\infty$ satisfy the same j -compatibility as the e_∞ : we may therefore normalise ζ_∞ (rescaling Θ_Π and $\Omega_{\tilde{\Pi}}^-$ accordingly) so that

$$\tilde{e}_\infty(\zeta_\infty \times \eta_\infty, -j) = e_\infty(\Pi_\infty \times \eta_\infty, -j) \quad \text{for all } 0 \leq j \leq a. \quad (28)$$

9 The p -adic L -function

Definition 9.0.1 (Test data). Fix the following data:

- at each $v \nmid p\infty$: a finite set I_v and collections $W_{v,i} \in \mathcal{W}(\Pi_v, E)$, $\Phi_{v,i} \in \mathcal{S}(\mathbb{Q}_v^2, \mathbb{Z})$, $i \in I_v$, with $\sum_i \tilde{Z}(W_{v,i}, \Phi_{v,i}; 1, \hat{\omega}_{\Pi,v} \hat{\eta}_{2,v}, s_1, s_2) = 1$ identically (Theorem 8.2.1); when Π_v , $\omega_{\Pi,v}$ and $\eta_{2,v}$ are unramified we take $|I_v| = 1$ with normalised spherical data (so $I := \prod_v I_v$ is finite). Note that the identity $\sum_i \tilde{Z} = 1$ persists under twisting χ_1, χ_2 by unramified characters, so applies with $(\chi_1, \chi_2) = (\hat{\eta}_v, \hat{\omega}_{\Pi,v} \hat{\eta}_v \hat{\eta}_{2,v})$ for η of p -power conductor;
- at p : φ_p with $\mathcal{W}(\varphi_p) = W^\alpha$ (Proposition 3.3.9);
- this yields vectors $\varphi_{f,i} \in \Pi_f$ and $\Phi_{f,i}^{(p)} \in \mathcal{S}(\mathbb{A}_f^{(p),2}, \mathbb{Z})$ for $i \in I$; fix a prime-to- p level $\mathcal{U}^{G,(p)}$ fixing all of them, and normalise Θ_Π so that all $\phi_{\varphi_{f,i}}$ are integral as in Section 4.8.

Definition 9.0.2. Let $\tilde{\Pi}$ be a regular, unramified P_1 -refined RACAR of weight $(a, 0, -a)$, and slope $h < a + 1$ and let ι denote the involution of $\mathcal{D}_h(\mathbb{Z}_p^\times, L)$ induced by $x \mapsto x^{-1}$ on $\mathbb{Z}_p^\times$. The *left-half p -adic L -function* of $\tilde{\Pi}$ is

$$L_p^-(\tilde{\Pi}) := \iota \left(\sum_{i \in I} \mu_{\tilde{\Pi}} \left(\phi_{\varphi_{f,i}}, \Phi_{f,i}^{(p)} \right) \right) \in \mathcal{D}_h(\mathbb{Z}_p^\times, L).$$

Remark 9.0.3. Following [LW26], the involution ι is included so that the interpolation property holds at the integers $\{0, -1, \dots, -a\}$ rather than $\{0, 1, \dots, a\}$, matching the critical range of the complex L -function. Note that

$$\int \eta^{-1}(x) x^{-j} (\iota \mu)(x) = \int \eta(x) x^j \mu(x).$$

Theorem 9.0.4. The distribution $L_p^-(\tilde{\Pi}) \in \mathcal{D}_h(\mathbb{Z}_p^\times, L)$ satisfies: for all $(-j, \eta) \in \text{Crit}_p^-(\Pi)$,

$$\int_{\mathbb{Z}_p^\times} \eta^{-1}(x) x^{-j} L_p^-(\tilde{\Pi})(x) = e_\infty(\Pi_\infty \times \eta_\infty, -j) e_p(\Pi_p \times \eta_p, -j) \cdot \frac{L^{(p)}(\Pi \times \eta, -j)}{\Omega_{\tilde{\Pi}}}.$$

Proof. Combine (26) (summed over the test data $i \in I$, which trivialises the local integrals away from $p\infty$) with (28) and the preceding remark. $\square$

Remark 9.0.5. The interpolation formula determines the values of $L_p^-(\tilde{\Pi})$ only at (j, η) with $(-1)^j = \omega_\Pi \eta(-1)$. As in [LW26, Conj. 1.2] one can further show the vanishing $\int \eta^{-1}(x) x^{-j} L_p^-(\tilde{\Pi}) = 0$ for characters of the opposite sign: the computation of Proposition 8.1.2 applies to every (j, η) , and for the wrong sign the archimedean integral vanishes identically.

Proposition 9.0.6. $L_p^-(\tilde{\Pi})$ is the unique element of $\mathcal{D}_h(\mathbb{Z}_p^\times, L)$ satisfying the interpolation property of Theorem 9.0.4 together with the vanishing at characters of the opposite sign.

Proof. The difference of two such elements lies in some $\mathcal{D}_{h'}(\mathbb{Z}_p^\times, L)$ with $h' < a + 1$ and their image under ι integrates to zero against $\eta(x) x^j$ for all finite-order η of p -power conductor and $0 \leq j \leq a$. The results then follows from Proposition 6.1.2(1) (and ι -invariance of $\mathcal{D}_h(\mathbb{Z}_p^\times, L)$). $\square$

Remark 9.0.7. If $\tilde{\Pi}$ is ordinary ($h = 0$), then $L_p^-(\tilde{\Pi})$ is a bounded measure and, by Proposition 9.0.6 applied with $h' = 0$, coincides with the p -adic L -function of [LW26, Definition 9.3] – including the choice of periods, provided the same normalisations of ζ_∞ and test data are made.

10 Duality and the right half of the critical strip

Finally we treat P_2 -refinements and the critical values $\text{Crit}_p^+(\Pi)$, by reduction to the P_1 -theory for the dual representation. Note that if Π is a RACAR of weight $(a, 0, -a)$ then so is $\Pi^\vee$, with $\omega_{\Pi^\vee} = \omega_\Pi^{-1}$ and $L(\Pi^\vee \times \eta^{-1}, s) = L((\Pi \times \eta)^\vee, s)$.

Lemma 10.0.1. $\sigma_p \times \sigma'_p$ is a regular P_2 -refinement of Π_p if and only if $(\sigma'_p)^\vee \times \sigma_p^\vee$ is a regular P_1 -refinement of $\Pi_p^\vee$, and the two refinements have the same slope. In particular, small slope P_2 -refinements of Π correspond to small slope P_1 -refinements of $\Pi^\vee$.

Proof. By combining the equality $w_0 \bar{P}_2 w_0^{-1} = P_1$ where w_0 is the longest Weyl element $w_0 = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} = w_0^{-1}$ with Casselman duality $J_{P_i}(\Pi_p^\vee) \cong J_{\bar{P}_i}(\Pi_p)^\vee$, we get an isomorphism $J_{P_1}(\Pi_p^\vee) \cong J_{P_2}(\Pi_p)^\vee \circ \text{Ad}(w_0)$. Under this isomorphism, a refinement $\sigma_p \times \sigma'_p$ of $J_{P_2}(\Pi_p)$ corresponds to a refinement $\sigma_p'^\vee \times \sigma_p^\vee$ (and notice here that central characters are inverted). Since $J_{P_2}(\Pi_p)$ has finite length and $\sigma_p \times \sigma'_p$ is regular (and is thus a direct summand), we can dualise (which is exact here), we get that $\sigma_p'^\vee \times \sigma_p^\vee$ is a direct summand of $J_{P_1}(\Pi_p^\vee)$ and thus a regular refinement (inverting central characters keeps them distinct). The converse direction follows similarly by $(\Pi_p^\vee, P_1)$ in place of (Π_p, P_2) and using $w_0 \bar{P}_1 w_0^{-1} = P_2$. Then

$$v_p(\sigma'^\vee(p)) = -v_p(\sigma'_p(p)) = v_p(\omega_{\sigma_p}(p)).$$

$\square$

Definition 10.0.2. Let $\tilde{\Pi} = (\Pi, \sigma_p \times \sigma'_p)$ be a regular small slope P_2 -refined RACAR of weight $(a, 0, -a)$, with σ'_p unramified, and let $\tilde{\Pi}^\vee$ denote the corresponding P_1 -refinement of $\Pi^\vee$ from Lemma 10.0.1. For $C \in \mathbb{Z}_p^\times$ let $[C] \in \mathcal{D}_0(\mathbb{Z}_p^\times, \mathcal{O})$ denote the Dirac measure at C , and recall the operators ι and tw_1 of Section 6 ($\text{tw}_1[C] = C[C]$), which preserve $\mathcal{D}_h(\mathbb{Z}_p^\times, L)$. The *right-half p -adic L -function* of $\tilde{\Pi}$ is

$$L_p^+(\tilde{\Pi}) := \frac{1}{\varepsilon^{(p)}(\Pi^{\vee, (p)}, 0)} \cdot (\iota \circ \text{tw}_1) \left(\left[N_\Pi^{(p)} \right] * L_p^-(\tilde{\Pi}^\vee) \right) \in \mathcal{D}_h(\mathbb{Z}_p^\times, L),$$

where $\varepsilon^{(p)}(\Pi^{\vee, (p)}, 0) = \prod_{\ell \neq p} \varepsilon_\ell(\Pi_\ell^\vee, 0)$ and $N_\Pi^{(p)}$ is the prime-to- p conductor, and $*$ is convolution of distributions. We also set $\Omega_\Pi^+ := \Omega_{\Pi^\vee}^-$.

Remark 10.0.3. Note that without loss of generality (after a twist as in Section 3.3), we may assume the character σ'_p , equivalently the P_1 -refinement $(\sigma'_p)^\vee$ of $\Pi_p^\vee$, is unramified. Indeed, let η be a Dirichlet character of p -power conductor such that $\sigma'_p \hat{\eta}_p$ is unramified. Then $(\sigma_p \otimes (\hat{\eta}_p \circ \det)) \times \sigma'_p \hat{\eta}_p$ is a P_2 -refinement of $(\Pi \times \eta)_p$. It is regular if and only if $\sigma_p \times \sigma'_p$ is, and it has the same slope since $\hat{\eta}_p(p) = 1$.

Proposition 10.0.4. *For all $(j+1, \eta) \in \text{Crit}_p^+(\Pi)$ we have*

$$\int_{\mathbb{Z}_p^\times} \eta(x) x^{j+1} L_p^+(\tilde{\Pi})(x) = e_\infty(\Pi_\infty \times \eta_\infty^{-1}, j+1) e_p(\Pi_p \times \eta_p^{-1}, j+1) \cdot \frac{L^{(p)}(\Pi \times \eta^{-1}, j+1)}{\Omega_\Pi^+}.$$

Proof. Identical to [LW26, Proposition 10.2], which we recall. Unwinding Definition 10.0.2 and applying Theorem 9.0.4 for $\tilde{\Pi}^\vee$,

$$\begin{aligned} \int \eta(x) x^{j+1} L_p^+(\tilde{\Pi}) &= \left(N_\Pi^{(p)} \right)^{-j} \eta \left(N_\Pi^{(p)} \right)^{-1} \varepsilon^{(p)}(\Pi^{\vee, (p)}, 0)^{-1} \\ &\quad \times e_\infty(\Pi_\infty^\vee \times \eta_\infty, -j) e_p(\Pi_p^\vee \times \eta_p, -j) \frac{L^{(p)}(\Pi^\vee \times \eta, -j)}{\Omega_{\Pi^\vee}^-}. \end{aligned}$$

By the identity of Coates [Coa89, (20)] cf. Section 3.4 and Lemma 2.18 of [LW26], the right-hand side equals

$$\begin{aligned} &\left(N_\Pi^{(p)} \right)^{-j} \eta \left(N_\Pi^{(p)} \right)^{-1} \varepsilon^{(p)}(\Pi^{\vee, (p)}, 0)^{-1} \left(\prod_{\ell \neq p} \varepsilon_\ell(\Pi_\ell^\vee \times \eta_\ell, -j) \right) \\ &\quad \times e_\infty(\Pi_\infty \times \eta_\infty^{-1}, j+1) e_p(\Pi_p \times \eta_p^{-1}, j+1) \frac{L^{(p)}(\Pi \times \eta^{-1}, j+1)}{\Omega_\Pi^+}, \end{aligned}$$

and since η is unramified at $\ell \neq p$, Tate's twisting formula for local ε -factors gives $\varepsilon_\ell(\Pi_\ell^\vee \times \eta_\ell, -j) = \varepsilon_\ell(\Pi_\ell^\vee, 0) \cdot \ell^{j c_\ell} \hat{\eta}_\ell(\ell^{c_\ell})$, with c_ℓ the exponent of ℓ in N_Π . Taking the product over $\ell \neq p$ and using the conventions of Section 3.1 to convert $\hat{\eta}^{(p)}(N_\Pi^{(p)}) = \eta(N_\Pi^{(p)})$, all factors to the left of e_∞ cancel and the proposition follows. $\square$

This completes the proofs of Theorems A and B of the introduction: Theorem A is Theorem 9.0.4 together with Proposition 9.0.6 and Remark 9.0.7, and Theorem B is Proposition 10.0.4.

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